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Optimal Economic Scheduling of Four
Electrical Generators Using CasADi and
IPOPT

Technical Report

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1 Introduction

The efficient and reliable operation of an electrical power system requires proper scheduling of the available generating units. When multiple generators are connected to a power system, the total electrical demand must be distributed among them in such a way that the required load is supplied while keeping the overall generation cost as low as possible.

In this work, an **economic scheduling problem involving four electrical generators** is considered over a **10-hour planning horizon**, with the time index given by

$$k = 0, 1, \dots, 10 \quad (1)$$

The power outputs of the four generators are treated as decision variables and are optimized at each time step.

The main objective is to **minimize the total generation cost** of all four generators over the complete planning horizon. Since each generator has a different nonlinear cost function, the optimal solution determines how much power should be generated by each unit at every hour.

At the same time, the total generation must satisfy the electrical demand at every time step. Therefore, the outputs of the four generators must satisfy the power-balance condition

$$x_1(k) + x_2(k) + x_3(k) + x_4(k) = d_k \quad (2)$$

In addition to demand satisfaction, each generator has a specified **minimum and maximum generation limit**. These limits ensure that the generators operate only within their permissible operating ranges.

The generators are also subject to a **ramp-rate constraint**, which limits how quickly their power output can change between consecutive hours. In this problem, the maximum allowable change in generation is **160 MW**. The same constraint is applied to the transition from the known generation at $k = -1$ to the scheduled generation at $k = 0$.

Ramp-Rate Constraint

The maximum allowable change in the power output of each generator between consecutive time steps is

$$|x_i(k) - x_i(k-1)| \leq 160 \text{ MW}$$

To formulate and solve this constrained nonlinear optimization problem, **CasADi** is used with its **Opti** framework, while **IPOPT** is used as the optimization solver. The resulting optimal generator schedules are then analyzed using the generation-demand profile, generator operating limits, and ramp-rate plots.

Optimization Framework

- ▶ **CasADi** is used to formulate the constrained nonlinear optimization problem.
- ▶ The **Opti** framework is used to define the optimization variables, objective function, and constraints.
- ▶ **IPOPT** is used to solve the resulting nonlinear optimization problem.

Thus, the overall objective of this work is to obtain an **economical and feasible generation schedule for the four generators** that minimizes the total generation cost while satisfying the demand, generation limits, and ramp-rate constraints.

Overall Objective

Obtain an economical and feasible generation schedule that

minimizes total generation cost,
satisfies the electrical demand,
respects generator operating limits,
and satisfies ramp-rate constraints.

The assignment requires these conditions to be demonstrated through the corresponding simulation results and plots.

2 Problem Statement

The problem considered in this work is the **optimal economic scheduling of four electrical generators over a 10-hour planning horizon**. The objective is to determine the generation setpoints of all four generators at each hour such that the total generation cost is minimized while satisfying the required operating constraints. The problem considers the hour indices

$$k = 0, 1, \dots, 10 \quad (3)$$

Let

$$x_k^{(i)} \in \mathbb{R} \quad (4)$$

denote the power output of generator i , measured in MW, at hour k , where

$$i \in \{1, 2, 3, 4\}. \quad (5)$$

Thus, at every hour, four generation setpoints have to be determined:

$$\boxed{x_k^{(1)}, \quad x_k^{(2)}, \quad x_k^{(3)}, \quad x_k^{(4)}} \quad (6)$$

The scheduler must determine these setpoints **before hour 0 begins**, using the known demand profile and the operating constraints of the generators.

2.1 Generator Cost

Each generator has an associated generation cost that depends on its power output. The cost function of generator i is given by a nonlinear function containing constant, linear, quadratic, and sinusoidal terms:

$$\boxed{c_i(x) = a_i + b_i x + c_i x^2 + d_i \sin\left(\frac{x}{d_i} + \theta_i\right)} \quad (7)$$

The parameters of the four generators are different, resulting in different generation costs for each unit.

Therefore, the scheduler has to decide how the total demand should be distributed among the four generators so that the **overall generation cost is as small as possible**.

Generator Cost

Each generator has a different nonlinear cost function. The optimization algorithm determines the generation allocation among the four units so that the total generation cost is minimized.

2.2 Demand Satisfaction

The first requirement is that the total power generated by the four generators must exactly satisfy the system demand at every hour. If d_k represents the demand at hour k , then

$$\boxed{x_k^{(1)} + x_k^{(2)} + x_k^{(3)} + x_k^{(4)} = d_k} \quad (8)$$

for

$$k = 0, 1, \dots, 10. \quad (9)$$

This ensures that there is no generation-demand mismatch at any time step. The demand values are provided in the `demand.xlsx` file.

Power-Balance Requirement

At every hour, the sum of the power outputs of the four generators must be exactly equal to the electrical demand.

$$\sum_{i=1}^4 x_k^{(i)} = d_k$$

2.3 Generation Limits

Each generator has a specified minimum and maximum allowable generation level. Hence, the output of every generator must remain within its permissible operating range:

$$x_{\min}^{(i)} \leq x_k^{(i)} \leq x_{\max}^{(i)} \quad (10)$$

for

$$i = 1, 2, 3, 4, \quad k = 0, 1, \dots, 10. \quad (11)$$

These limits prevent the optimization algorithm from assigning an infeasible generation level to any generator.

The generation limits used in the MATLAB implementation are shown in Table 1.

Table 1: Minimum and maximum generation limits of the four generators

Generator	Minimum (MW)	Maximum (MW)
G_1	100	950
G_2	120	1000
G_3	150	900
G_4	100	980

The corresponding constraints can therefore be written as

$$100 \leq x_k^{(1)} \leq 950, \quad (12)$$

$$120 \leq x_k^{(2)} \leq 1000, \quad (13)$$

$$150 \leq x_k^{(3)} \leq 900, \quad (14)$$

$$100 \leq x_k^{(4)} \leq 980. \quad (15)$$

These constraints must hold for every

$$k = 0, 1, \dots, 10.$$

2.4 Ramp-Rate Constraint

The generation output of a generator cannot change by an unlimited amount between two consecutive hours. Therefore, the change in generation is restricted by the ramp-rate limit u :

$$\boxed{\left| x_{k+1}^{(i)} - x_k^{(i)} \right| \leq u} \quad (16)$$

where

$$\boxed{u = 160 \text{ MW}.} \quad (17)$$

This constraint is imposed for all four generators throughout the planning horizon.

The transition from the known generation at $k = -1$ to the generation at $k = 0$ must also satisfy the same limit:

$$\boxed{\left| x_0^{(i)} - x_{-1}^{(i)} \right| \leq 160 \text{ MW}} \quad (18)$$

The known previous-hour outputs are

$$x_{-1}^{(1)} = 724.4 \text{ MW}, \quad (19)$$

$$x_{-1}^{(2)} = 865.9 \text{ MW}, \quad (20)$$

$$x_{-1}^{(3)} = 819.5 \text{ MW}, \quad (21)$$

$$x_{-1}^{(4)} = 783.7 \text{ MW}. \quad (22)$$

Ramp-Rate Requirement

The power output of each generator cannot increase or decrease by more than **160 MW** between consecutive hours. The same restriction is applied to the initial transition from $k = -1$ to $k = 0$.

2.5 Overall Optimization Objective

The scheduler must determine the setpoints of all four generators for $k = 0, \dots, 10$ such that the **total cost over all generators and all hours is minimized**, while satisfying all the constraints described above.

Thus, the optimization problem can be summarized as

$$\boxed{\min_{\{x_k^{(i)}\}} \sum_{k=0}^{10} \sum_{i=1}^4 c_i \left(x_k^{(i)} \right)} \quad (23)$$

subject to the demand-supply equality constraint

$$x_k^{(1)} + x_k^{(2)} + x_k^{(3)} + x_k^{(4)} = d_k \quad (24)$$

the generation-limit constraints

$$x_{\min}^{(i)} \leq x_k^{(i)} \leq x_{\max}^{(i)} \quad (25)$$

and the ramp-rate constraints

$$|x_{k+1}^{(i)} - x_k^{(i)}| \leq 160 \text{ MW}. \quad (26)$$

Complete Optimization Objective

Determine the optimal generation setpoints of the four generators that minimize the total generation cost while simultaneously satisfying:

- ▶ Demand-supply equality
- ▶ Generator minimum and maximum limits
- ▶ Ramp-rate constraints

The resulting optimization problem is then formulated in **CasADi** using the **Opti framework** and solved using **IPOPT**.

3 Mathematical Formulation of the Optimization Problem

The economic scheduling problem is formulated as a **constrained nonlinear optimization problem**. The purpose of the formulation is to determine the power output of each of the four generators at every hour of the planning horizon such that the total generation cost is minimized while satisfying the required demand and generator operating constraints.

The mathematical formulation consists of four main components:

1. Decision variables
2. Generator cost functions
3. Objective function
4. Operating constraints

3.1 Decision Variables

The primary decision variables in the optimization problem are the **power generation setpoints of the four generators**.

Let

$$x_i(k) \tag{27}$$

denote the power generated by generator i at hour k , where

$$i = 1, 2, 3, 4 \tag{28}$$

and

$$k = 0, 1, \dots, 10. \tag{29}$$

The unit of $x_i(k)$ is MW.

Therefore, the four generator outputs at each time step are

$$\boxed{x_1(k), \quad x_2(k), \quad x_3(k), \quad x_4(k)} \tag{30}$$

Since there are 11 time points and four generators, the optimization determines a total of

$$\boxed{4 \times 11 = 44} \tag{31}$$

generation setpoints.

The complete decision-variable vectors can be represented as

$$\mathbf{x}_1 = [x_1(0) \quad x_1(1) \quad \cdots \quad x_1(10)], \tag{32}$$

$$\mathbf{x}_2 = [x_2(0) \quad x_2(1) \quad \cdots \quad x_2(10)], \tag{33}$$

$$\mathbf{x}_3 = [x_3(0) \quad x_3(1) \quad \cdots \quad x_3(10)], \tag{34}$$

and

$$\mathbf{x}_4 = [x_4(0) \quad x_4(1) \quad \cdots \quad x_4(10)]. \tag{35}$$

Thus, the optimization algorithm determines the complete generation trajectory of each generator over the 10-hour planning horizon. The assignment defines $x_k^{(i)}$ as the power output of generator i at hour k .

Decision Variables

The 44 optimization variables represent the generation setpoints of four generators over 11 time points:

$$\mathbf{x} = [\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \mathbf{x}_4]$$

3.2 Generator Cost Functions

Each generator has a different operating cost associated with its power output. Therefore, the cost of generation is represented as a function of the corresponding generator setpoint.

The general nonlinear cost function specified for generator i is

$$C_i(x) = a_i + b_i x + c_i x^2 + d_i \sin\left(\frac{x}{d_i} + \theta_i\right) \quad (36)$$

where a_i , b_i , c_i , d_i , and θ_i are parameters associated with generator i .

The cost function contains several terms:

- ▶ a_i represents a constant cost component.
- ▶ $b_i x$ represents the linear dependence of cost on generated power.
- ▶ $c_i x^2$ represents the quadratic component of the generation cost.
- ▶ The sinusoidal term introduces additional nonlinear variation in the cost.

The assignment specifies that the four generators have different values of these parameters, resulting in different cost characteristics for each generator.

3.2.1 Generator 1

For generator 1, the cost function implemented in MATLAB is

$$C_1(x_1) = 100 + 2.5x_1 + 0.0055x_1^2 + 50 \sin\left(\frac{x_1}{120}\right) \quad (37)$$

where x_1 is the power output of generator 1 in MW.

Generator 1 Cost Parameters

$$a_1 = 100, \quad b_1 = 2.5, \quad c_1 = 0.0055$$

with sinusoidal amplitude 50 and scaling factor 120.

3.2.2 Generator 2

For generator 2, the cost function is

$$C_2(x_2) = 120 + 2.2x_2 + 0.0038x_2^2 + 40 \sin\left(\frac{x_2}{140} + 0.5\right) \quad (38)$$

where x_2 represents the power output of generator 2.

Generator 2 Cost Parameters

$$a_2 = 120, \quad b_2 = 2.2, \quad c_2 = 0.0038$$

with sinusoidal amplitude 40, scaling factor 140, and phase shift 0.5.

3.2.3 Generator 3

For generator 3, the cost function is

$$C_3(x_3) = 200 + 2.8x_3 + 0.0022x_3^2 + 60 \sin\left(\frac{x_3}{110} - 0.3\right) \quad (39)$$

where x_3 represents the power output of generator 3.

Generator 3 Cost Parameters

$$a_3 = 200, \quad b_3 = 2.8, \quad c_3 = 0.0022$$

with sinusoidal amplitude 60, scaling factor 110, and phase shift -0.3 .

3.2.4 Generator 4

For generator 4, the cost function is

$$C_4(x_4) = 400 + 3.2x_4 + 0.0015x_4^2 + 45 \sin\left(\frac{x_4}{130} + 1.0\right) \quad (40)$$

where x_4 represents the power output of generator 4.

Generator 4 Cost Parameters

$$a_4 = 400, \quad b_4 = 3.2, \quad c_4 = 0.0015$$

with sinusoidal amplitude 45, scaling factor 130, and phase shift 1.0.

3.2.5 Significance of the Cost Functions

The four cost functions are different, meaning that the generators do not have the same cost characteristics. For a particular amount of generated power, one generator may be more economical than another. Therefore, the optimizer must determine the appropriate distribution of generation among the four generators.

The sinusoidal terms make the cost functions **nonlinear** and introduce variations in the cost characteristics with respect to generator output. Consequently, the overall scheduling problem is a nonlinear optimization problem rather than a simple linear or purely quadratic optimization problem.

The objective of the optimization is therefore not to minimize the cost of any individual generator separately, but to determine the combination of $x_1(k)$, $x_2(k)$, $x_3(k)$, and $x_4(k)$ that results in the **minimum total generation cost** while satisfying all system constraints.

The cost functions used in the optimization are directly implemented in the MATLAB code through the summation of the individual generator costs over all time steps.

Key Point

The optimizer determines the generation combination of all four generators that gives the lowest total generation cost while satisfying the demand and all operating constraints.

4 Objective Function

The main objective of the generator scheduling problem is to determine the power output of the four generators such that the **total generation cost over the entire planning horizon is minimized**. Since the planning horizon consists of the time points $k = 0, 1, \dots, 10$, the optimization considers **11 generation points** for each generator.

For each generator, the generation cost depends on its power output. The four generators have different nonlinear cost functions, represented by $C_1(x_1)$, $C_2(x_2)$, $C_3(x_3)$, and $C_4(x_4)$. Therefore, the cost incurred at a particular time step k is

$$C_1(x_1(k)) + C_2(x_2(k)) + C_3(x_3(k)) + C_4(x_4(k)). \quad (41)$$

The total cost over the complete planning horizon is obtained by summing this cost over all the time steps:

$$J = \sum_{k=0}^{10} [C_1(x_1(k)) + C_2(x_2(k)) + C_3(x_3(k)) + C_4(x_4(k))]$$
 (42)

where

- J represents the total generation cost over the planning horizon.

- $x_i(k)$ represents the power output of generator i at time step k .
- $C_i(x_i(k))$ represents the generation cost of generator i at time step k .
- $k = 0, \dots, 10$ represents the 11 time points in the planning horizon.

Thus, the optimization problem attempts to find the values of

$$\boxed{x_1(k), \quad x_2(k), \quad x_3(k), \quad x_4(k)} \quad (43)$$

for every k that result in the lowest possible value of J .

4.1 Implementation of the Objective Function in MATLAB

The objective function used in the MATLAB code is implemented by first initializing the individual costs:

```
c1 = 0;
c2 = 0;
c3 = 0;
c4 = 0;
```

The cost of each generator is then accumulated over all 11 time points using a **for** loop:

```
for k = 1:N+1
```

```
    c1 = c1 + 100 + 2.5*x1(k)
        + 0.0055*x1(k)^2
        + 50*sin(x1(k)/120);

    c2 = c2 + 120 + 2.2*x2(k)
        + 0.0038*x2(k)^2
        + 40*sin(x2(k)/140 + 0.5);

    c3 = c3 + 200 + 2.8*x3(k)
        + 0.0022*x3(k)^2
        + 60*sin(x3(k)/110 - 0.3);

    c4 = c4 + 400 + 3.2*x4(k)
        + 0.0015*x4(k)^2
        + 45*sin(x4(k)/130 + 1.0);
```

```
end
```

Here, c_1 , c_2 , c_3 , and c_4 represent the accumulated generation costs of the four generators over the complete planning horizon.

The four accumulated costs are then combined to form the overall objective function:

```
optimizer.minimize(c1+c2+c3+c4);
```


Therefore, the optimization objective can be expressed as

$$\boxed{\min J} \quad (44)$$

where J represents the total generation cost over all four generators and all 11 time points.

Objective Function in the Optimization Framework

The total objective passed to the CasADi optimizer is the sum of the four accumulated generator costs:

$$\boxed{J = c_1 + c_2 + c_3 + c_4}$$

The optimizer then minimizes this total cost subject to all the system constraints.

4.2 Constrained Minimization

It is important to note that the objective function is minimized **subject to the system constraints**. The optimizer cannot simply reduce the generation cost by selecting arbitrary generator outputs.

The selected generation must simultaneously satisfy the following conditions:

1. The total generation must satisfy the demand-supply equality.
2. Each generator must operate within its specified minimum and maximum generation limits.
3. The change in generator output between consecutive hours must satisfy the ramp-rate constraint.

Therefore, the optimization does not seek merely the minimum possible generation cost. Instead, it seeks the

Minimum-Cost Feasible Generation Schedule

This means that the generation schedule must be both **economical** and **feasible**.

4.3 Economic Interpretation

The objective function determines the economic performance of the generator scheduling problem. Since the four generators have different nonlinear cost characteristics, the optimizer searches for the most economical combination of generator outputs at every time step.

The optimization therefore determines how the required electrical demand should be distributed among the four generators while considering their individual generation costs and operating constraints.

In summary, the objective function can be represented as

$$\min_{\{x_i(k)\}} \sum_{k=0}^{10} \sum_{i=1}^4 C_i(x_i(k)) \quad (45)$$

subject to all demand, generation-limit, and ramp-rate constraints.

Key Result

The objective function provides the economic criterion for the optimization. The optimal solution is the combination of generator outputs that supplies the required electrical demand at the **lowest total generation cost over the 10-hour planning horizon**, while satisfying all specified operating constraints.

5 Constraints

The optimization problem is subject to several constraints that ensure the obtained generator schedule is both **economically optimal and physically feasible**. Although the objective function determines the least-cost operating schedule, the generator outputs cannot be selected arbitrarily. At every hour, the total generation must satisfy the system demand, each generator must operate within its allowable generation range, and the change in generator output between consecutive hours must remain within the specified ramp-rate limit.

The three main constraints considered in this problem are:

1. Demand-supply equality constraint
2. Generator generation-limit constraint
3. Ramp-rate constraint

5.1 Demand-Supply Equality Constraint

The most important constraint in the generator scheduling problem is the requirement that the total power generated by all four generators must exactly match the electrical demand at every time step.

Let d_k denote the electrical demand at hour k . The total generation is the sum of the outputs of the four generators:

$$P_{\text{total}}(k) = x_1(k) + x_2(k) + x_3(k) + x_4(k). \quad (46)$$

Therefore, the power-balance constraint is given by

$$x_1(k) + x_2(k) + x_3(k) + x_4(k) = d_k \quad (47)$$

for

$$k = 0, 1, \dots, 10. \quad (48)$$

This is an **equality constraint**, meaning that the total generation must be exactly equal to the specified demand rather than simply being greater than or less than the demand. The assignment explicitly requires this demand-supply equality at every time step.

The demand profile used in the optimization is given in Table 2.

Table 2: Demand profile used in the optimization

Hour k	Demand d_k (MW)
0	2929.83
1	3350.07
2	3537.73
3	3466.92
4	3394.74
5	3022.01
6	2938.21
7	2443.04
8	2318.16
9	2949.45
10	3253.15

For example, at $k = 0$,

$$x_1(0) + x_2(0) + x_3(0) + x_4(0) = 2929.83 \text{ MW} \quad (49)$$

Similarly, at $k = 2$,

$$x_1(2) + x_2(2) + x_3(2) + x_4(2) = 3537.73 \text{ MW} \quad (50)$$

Thus, as the demand changes from one hour to another, the optimizer must appropriately redistribute the generation among the four generators.

In the MATLAB implementation, this condition is imposed using:

```
optimizer.subject_to( ...
    x1(k)+x2(k)+x3(k)+x4(k)==demand(k) )
```

Hence, the optimized solution must always maintain the required **generation-demand balance**.

5.2 Generation Limits

Each generator has a minimum and maximum power generation capability. Therefore, the output of each generator must remain within its specified operating range at every

time step.

The generation-limit constraint is generally written as

$$\boxed{x_i^{\min} \leq x_i(k) \leq x_i^{\max}} \quad (51)$$

for

$$i = 1, 2, 3, 4 \quad (52)$$

and

$$k = 0, 1, \dots, 10. \quad (53)$$

The limits used in the present problem are shown in Table 3.

Table 3: Generation limits of the four generators

Generator	Lower Limit (MW)	Upper Limit (MW)
G_1	100	950
G_2	120	1000
G_3	150	900
G_4	100	980

Therefore, the individual constraints are

$$\boxed{100 \leq x_1(k) \leq 950} \quad (54)$$

$$\boxed{120 \leq x_2(k) \leq 1000} \quad (55)$$

$$\boxed{150 \leq x_3(k) \leq 900} \quad (56)$$

and

$$\boxed{100 \leq x_4(k) \leq 980} \quad (57)$$

for all

$$k = 0, 1, \dots, 10.$$

These constraints must hold for every value of k . The assignment specifies that every generator setpoint must remain between its corresponding lower and upper generation bounds.

The lower limits ensure that a generator does not operate below its permitted minimum output, while the upper limits prevent the optimizer from assigning more power to a generator than its maximum available generation.

In MATLAB, these constraints are implemented as:

```
optimizer.subject_to(lb1 <= x1 <= ub1);
optimizer.subject_to(lb2 <= x2 <= ub2);
optimizer.subject_to(lb3 <= x3 <= ub3);
optimizer.subject_to(lb4 <= x4 <= ub4);
```

These constraints restrict the feasible region of the optimization problem and ensure that the obtained generator schedule remains within the allowable operating range.

5.3 Ramp-Rate Constraints

The third constraint is associated with the **rate at which the generator output can change**. A generator cannot instantaneously increase or decrease its power output by an unlimited amount. Therefore, the difference between the generation levels at two consecutive time steps is restricted.

The ramp-rate constraint is given by

$$|x_i(k) - x_i(k-1)| \leq 160 \text{ MW} \quad (58)$$

for the four generators and for the applicable consecutive time steps.

The value of the maximum allowable ramp rate is

$$u = 160 \text{ MW} \quad (59)$$

The assignment specifies that the change in generation between consecutive hours must satisfy this limit.

This constraint can also be expressed as two inequalities:

$$x_i(k) - x_i(k-1) \leq 160 \quad (60)$$

and

$$x_i(k-1) - x_i(k) \leq 160. \quad (61)$$

Therefore,

$$-160 \leq x_i(k) - x_i(k-1) \leq 160 \quad (62)$$

This means that a generator can increase its output by at most **160 MW** or decrease its output by at most **160 MW** between two consecutive time steps.

For example, if

$$x_1(k-1) = 700 \text{ MW}, \quad (63)$$

then the next generation value must satisfy

$$\boxed{540 \leq x_1(k) \leq 860 \text{ MW}} \quad (64)$$

Thus, even if the cost minimization would prefer a much larger change, the ramp-rate constraint prevents such an abrupt change.

5.3.1 Initial Ramp-Rate Constraint

A special case occurs at the beginning of the planning horizon. The generation values at $k = -1$ are already known, and the optimizer must determine the generation at $k = 0$. Therefore, the transition from $k = -1$ to $k = 0$ must also satisfy the same 160 MW ramp-rate limit:

$$\boxed{|x_i(0) - x_i(-1)| \leq 160 \text{ MW}} \quad (65)$$

The known generation values at $k = -1$ are

$$x_1(-1) = 724.4 \text{ MW}, \quad (66)$$

$$x_2(-1) = 865.9 \text{ MW}, \quad (67)$$

$$x_3(-1) = 819.5 \text{ MW}, \quad (68)$$

$$x_4(-1) = 783.7 \text{ MW}. \quad (69)$$

Hence, for generator 1, for example,

$$\boxed{|x_1(0) - 724.4| \leq 160} \quad (70)$$

Similarly,

$$\boxed{|x_2(0) - 865.9| \leq 160} \quad (71)$$

$$\boxed{|x_3(0) - 819.5| \leq 160} \quad (72)$$

and

$$\boxed{|x_4(0) - 783.7| \leq 160} \quad (73)$$

The assignment explicitly requires this initial transition to satisfy the ramp-rate constraint as well.

In the MATLAB code, the ramp-rate constraints between consecutive time steps are implemented using:

```
optimizer.subject_to(abs(x1(k)-x1(k-1)) <= 160)
optimizer.subject_to(abs(x2(k)-x2(k-1)) <= 160)
optimizer.subject_to(abs(x3(k)-x3(k-1)) <= 160)
optimizer.subject_to(abs(x4(k)-x4(k-1)) <= 160)
```

The initial transition is separately imposed using:

```
optimizer.subject_to(abs(x1(1)-x1_init) <= 160)
optimizer.subject_to(abs(x2(1)-x2_init) <= 160)
optimizer.subject_to(abs(x3(1)-x3_init) <= 160)
optimizer.subject_to(abs(x4(1)-x4_init) <= 160)
```

Thus, the ramp-rate constraint ensures that the optimized generation schedule does not require unrealistic changes in generator output.

5.4 Summary of Constraints

The complete optimization problem is therefore subject to the following three types of constraints.

5.4.1 Demand-Supply Balance

$$\boxed{x_1(k) + x_2(k) + x_3(k) + x_4(k) = d_k} \quad (74)$$

5.4.2 Generation Limits

$$\boxed{x_i^{\min} \leq x_i(k) \leq x_i^{\max}} \quad (75)$$

5.4.3 Ramp-Rate Limits

$$\boxed{|x_i(k) - x_i(k-1)| \leq 160 \text{ MW}} \quad (76)$$

including the initial transition

$$\boxed{|x_i(0) - x_i(-1)| \leq 160 \text{ MW}} \quad (77)$$

Constraint Summary

The optimization solution must simultaneously satisfy:

$$\begin{aligned} &\text{Generation} = \text{Demand}, \\ &x_i^{\min} \leq x_i(k) \leq x_i^{\max}, \\ &|x_i(k) - x_i(k-1)| \leq 160 \text{ MW}. \end{aligned}$$

Together, these constraints ensure that the final solution is not only **economically optimal**, but also **feasible with respect to demand requirements and generator operating capabilities**.

6 Initial Conditions

Before the optimization starts, the power output of each generator at the previous time step, $k = -1$, is assumed to be known. These values represent the **initial operating condition of the four generators immediately before the scheduling horizon begins**.

The given initial generation values are:

$$x_1(-1) = 724.4 \text{ MW} \quad (78)$$

$$x_2(-1) = 865.9 \text{ MW} \quad (79)$$

$$x_3(-1) = 819.5 \text{ MW} \quad (80)$$

$$x_4(-1) = 783.7 \text{ MW} \quad (81)$$

These values are important because the optimization starts determining the new generator setpoints from $k = 0$. Therefore, the change from the existing operating point at $k = -1$ to the first scheduled operating point at $k = 0$ must also be physically feasible.

6.1 Initial Ramp-Rate Constraint

The initial values are used to impose the first **ramp-rate constraint**:

$$|x_i(0) - x_i(-1)| \leq 160 \text{ MW} \quad (82)$$

for

$$i = 1, 2, 3, 4. \quad (83)$$

For each generator, this gives:

$$|x_1(0) - 724.4| \leq 160 \quad (84)$$

$$|x_2(0) - 865.9| \leq 160 \quad (85)$$

$$|x_3(0) - 819.5| \leq 160 \quad (86)$$

$$|x_4(0) - 783.7| \leq 160 \quad (87)$$

Thus, the first scheduled generation values are restricted by the actual operating condition of the generators at the previous hour.

Initial Transition Requirement

The first scheduled operating point at $k = 0$ cannot be selected independently of the previous operating point. The transition from $k = -1$ to $k = 0$ must satisfy the same ramp-rate limit of **160 MW**.

6.2 Initial Feasible Ranges

For generator 1, for example,

$$|x_1(0) - 724.4| \leq 160 \quad (88)$$

can be written as

$$564.4 \leq x_1(0) \leq 884.4 \text{ MW} \quad (89)$$

Similarly, the initial feasible ranges obtained from the ramp-rate constraint are shown in Table 4.

Table 4: Initial generation values and feasible ranges for $x_i(0)$ based on the ramp-rate constraint

Generator	Previous Output $x_i(-1)$ (MW)	Allowed $x_i(0)$ (MW)
G_1	724.4	564.4–884.4
G_2	865.9	705.9–1025.9
G_3	819.5	659.5–979.5
G_4	783.7	623.7–943.7

These ranges are considered together with the generator minimum and maximum limits. Therefore, the optimizer cannot choose the first generation setpoints arbitrarily; they must satisfy **both the generator operating limits and the initial ramp-rate constraint**.

6.3 MATLAB Implementation

In the MATLAB implementation, the initial values are defined as:

```
x1_init = 724.4;
x2_init = 865.9;
x3_init = 819.5;
x4_init = 783.7;
```

The initial ramp-rate constraints are imposed using:

```
optimizer.subject_to(abs(x1(1)-x1_init) <= 160)
optimizer.subject_to(abs(x2(1)-x2_init) <= 160)
optimizer.subject_to(abs(x3(1)-x3_init) <= 160)
optimizer.subject_to(abs(x4(1)-x4_init) <= 160)
```

These constraints connect the known operating condition at $k = -1$ with the first optimized generation point at $k = 0$.

6.4 Significance of Initial Conditions

The initial conditions provide the **starting point for the optimization horizon**. They ensure that the first scheduled generation values are consistent with the actual operating condition of the generators immediately before the optimization horizon begins.

Without the initial ramp-rate constraints, the optimizer could potentially select a first-hour generation value that is within the generator's minimum and maximum limits but requires an unrealistically large change from the previous operating point.

Therefore, the initial conditions play an important role in ensuring that the complete generation schedule is physically feasible from the very beginning of the planning horizon.

Key Point

The initial generation values at $k = -1$ establish the starting operating condition. The optimizer must determine $x_i(0)$ such that the transition from $k = -1$ to $k = 0$ does not exceed the **160 MW ramp-rate limit**, while also satisfying the generator operating limits.

Hence, these initial conditions provide the **starting point for the optimization horizon** and ensure that the transition into the first scheduled hour is consistent with the physical ramping capability of each generator.

7 Demand Data

The electrical demand profile is a key input to the generator scheduling optimization problem. At every hour of the planning horizon, the total power generated by the four generators must be equal to the corresponding electrical demand. Therefore, the demand values determine the total amount of power that must be produced at each time step.

The planning horizon considered in this work consists of the hour indices

$$\boxed{k = 0, 1, \dots, 10} \quad (90)$$

Hence, the optimization requires **11 demand values**, one for each time step. The demand data used in the problem are given in MW as shown in Table 5.

Table 5: Electrical demand profile over the planning horizon

Hour k	Demand d_k (MW)
0	2929.83
1	3350.07
2	3537.73
3	3466.92
4	3394.74
5	3022.01
6	2938.21
7	2443.04
8	2318.16
9	2949.45
10	3253.15

Demand Profile

The demand values represent the required total electrical generation at each time step. The four generators must collectively produce exactly the corresponding demand at every hour.

7.1 Demand Variation Over the Planning Horizon

The demand is not constant throughout the planning horizon. It increases during the initial part of the horizon, reaches its highest value at

$$\boxed{k = 2} \quad (91)$$

where

$$\boxed{d_2 = 3537.73 \text{ MW}} \quad (92)$$

and then decreases significantly until reaching its lowest value at

$$\boxed{k = 8} \quad (93)$$

where

$$\boxed{d_8 = 2318.16 \text{ MW}.} \quad (94)$$

After $k = 8$, the demand increases again, reaching

$$\boxed{d_{10} = 3253.15 \text{ MW}} \quad (95)$$

at the final time step.

The overall variation can therefore be summarized as

Demand Variation

Initial increase $\rightarrow$ Maximum demand at $k = 2 \rightarrow$ Decrease $\rightarrow$
 Minimum demand at $k = 8 \rightarrow$ Final increase at $k = 10$

This variation in demand requires the generation outputs of the four generators to change accordingly. However, these changes cannot be made arbitrarily because the generator outputs are also restricted by their generation limits and the **160 MW ramp-rate constraint**.

7.2 Use of Demand Data in the Optimization

The demand values are stored in an Excel file named `demand.xlsx`. The MATLAB program reads the demand data using:

```
T = readtable('demand.xlsx');
demand = T.demand_MW;
```

Here, `T` stores the contents of the Excel table, while `demand` extracts the column containing the demand values in MW.

The value corresponding to each time step is then used in the demand-supply equality constraint:

$$\boxed{x_1(k) + x_2(k) + x_3(k) + x_4(k) = d_k} \quad (96)$$

For example, at hour $k = 0$,

$$\boxed{x_1(0) + x_2(0) + x_3(0) + x_4(0) = 2929.83 \text{ MW}} \quad (97)$$

while at hour $k = 2$,

$$x_1(2) + x_2(2) + x_3(2) + x_4(2) = 3537.73 \text{ MW} \quad (98)$$

Thus, the demand data directly determine the required **total generation at every time step**.

7.3 Time Vector

The MATLAB code also defines the time vector as

```
N = 10;
t = [0:1:N] ';
```

which produces the hour indices

$$t = [0, 1, 2, \dots, 10]^T \quad (99)$$

This time vector is subsequently used to plot the demand and the optimized generator outputs.

Role of the Time Vector

The time vector provides the 11 discrete time points corresponding to the scheduling horizon and is used as the horizontal axis for the demand and generator-output plots.

7.4 Role of the Demand Profile

The demand profile acts as the **time-varying load requirement** for the optimization problem. At every time step, the optimizer must determine the most economical combination of the four generator outputs such that their total output follows the specified demand.

Therefore, the generation scheduling process can be represented as

$$\text{Demand Profile} \longrightarrow \text{Optimization} \longrightarrow \text{Generator Setpoints} \quad (100)$$

where the generator setpoints must satisfy the demand while also respecting the generator operating limits and ramp-rate constraints.

In summary, the demand profile provides the required total power at each hour and directly influences the optimal generation schedule. The optimizer determines the most economical combination of the four generator outputs while ensuring that the total generation follows the demand profile at every time step.

Key Point

The demand profile is a fundamental input to the economic scheduling problem. The optimized outputs of the four generators must satisfy

$$x_1(k) + x_2(k) + x_3(k) + x_4(k) = d_k$$

at every time step $k = 0, \dots, 10$.

8 Optimization Methodology

The formulated generator scheduling problem is a **constrained nonlinear optimization problem** because the objective function contains nonlinear terms and the generator outputs are subject to several equality and inequality constraints. The optimization problem is implemented in MATLAB using the **CasADi** framework and solved using the **IPOPT** nonlinear programming solver.

The overall procedure followed in the MATLAB implementation is:

Define Optimization Problem → Define Decision Variables → Define Cost Functions
→ Define Objective → Add Constraints → Solve Using IPOPT
→ Extract Optimal Solution

(101)

8.1 CasADi Optimization Framework

CasADi is used to formulate the mathematical optimization problem in MATLAB. It provides tools for defining symbolic variables, nonlinear objective functions, and constraints, and then passing the resulting optimization problem to a suitable numerical solver.

In the present work, the **Opti** framework of CasADi is used to construct the optimization problem. The optimization environment is created using:

```
optimizer = casadi.Opti();
```

The **Opti** object acts as the main framework in which the decision variables, objective function, and constraints are defined.

The use of the **Opti** framework makes it possible to formulate the complete generator scheduling problem in a structured manner. The objective and constraints can be written directly using the optimization variables, allowing CasADi to construct the corresponding nonlinear optimization problem.

CasADi Opti Framework

The `Opti` framework provides the optimization environment in which the generator decision variables, objective function, and operating constraints are defined.

8.2 Definition of Decision Variables

After creating the optimization environment, the power outputs of the four generators are defined as optimization variables.

The MATLAB code uses:

```
x1 = optimizer.variable(1,N+1);  
x2 = optimizer.variable(1,N+1);  
x3 = optimizer.variable(1,N+1);  
x4 = optimizer.variable(1,N+1);
```

Since

$$N = 10, \quad (102)$$

each optimization variable contains

$$\boxed{N + 1 = 11} \quad (103)$$

elements corresponding to

$$\boxed{k = 0, 1, \dots, 10.} \quad (104)$$

Therefore,

$$\mathbf{x}_1 = \begin{bmatrix} x_1(0) & x_1(1) & \cdots & x_1(10) \end{bmatrix}, \quad (105)$$

and similarly for the other three generators.

Thus, the optimization algorithm simultaneously determines the generation trajectory of all four generators over the complete planning horizon.

In total, there are

$$\boxed{4 \times 11 = 44} \quad (106)$$

generator setpoints that are determined by the optimization.

8.3 Definition of Generator Limits

Before solving the optimization problem, the lower and upper generation limits of each generator are defined in the MATLAB code:

```
lb1 = 100;   ub1 = 950;
lb2 = 120;   ub2 = 1000;
lb3 = 150;   ub3 = 900;
lb4 = 100;   ub4 = 980;
```

These values are then imposed as constraints on the optimization variables.

For example,

```
optimizer.subject_to(lb1 <= x1 <= ub1);
```

ensures that every value of x_1 remains between 100 MW and 950 MW.

The same procedure is applied to the other three generators. Therefore, the optimizer is restricted to feasible generator operating ranges.

The generation limits can be summarized as

Table 6: Generator operating limits used in the optimization		
Generator	Lower Limit (MW)	Upper Limit (MW)
G_1	100	950
G_2	120	1000
G_3	150	900
G_4	100	980

8.4 Definition of Cost Functions

The next step is to construct the generation cost of each generator. The MATLAB code initializes the four cost variables:

```
c1 = 0;
c2 = 0;
c3 = 0;
c4 = 0;
```

A **for** loop is then used to calculate the generation cost at every time step:

```
for k = 1:N+1
```

Within this loop, the individual generator costs are added for every value of k .

For generator 1:

```
c1 = c1 + 100 + 2.5*x1(k)
      + 0.0055*x1(k)^2
      + 50*sin(x1(k)/120);
```

Similarly, the costs of generators 2, 3, and 4 are calculated using their respective nonlinear cost functions.

As a result, c_1 , c_2 , c_3 , and c_4 represent the accumulated generation costs of the four generators over the entire planning horizon.

8.5 Formation of the Objective Function

After calculating the individual costs, the total generation cost is formed by adding the costs of all four generators:

$$J = c_1 + c_2 + c_3 + c_4 \quad (107)$$

The optimization objective is then defined using:

```
optimizer.minimize(c1+c2+c3+c4);
```

Therefore, the solver attempts to find the values of the 44 decision variables that produce the **minimum possible total generation cost**.

Mathematically, the objective is

$$\min \sum_{k=0}^{10} [C_1(x_1(k)) + C_2(x_2(k)) + C_3(x_3(k)) + C_4(x_4(k))] \quad (108)$$

subject to all the specified constraints.

Objective Formation

The individual generation costs accumulated over all time steps are combined to form the total objective function. The optimization then minimizes this total cost while satisfying the operating constraints.

$$J = c_1 + c_2 + c_3 + c_4$$

8.6 Addition of Demand-Supply Constraints

Once the objective has been defined, the demand-supply equality constraint is imposed.

The MATLAB code uses:

```
for k = 1:N+1
    optimizer.subject_to(
        x1(k)+x2(k)+x3(k)+x4(k)==demand(k))
end
```

This ensures that at every time step,

$$x_1(k) + x_2(k) + x_3(k) + x_4(k) = d_k \quad (109)$$

Therefore, the optimizer cannot reduce the total generation simply to reduce the cost. It must generate exactly the amount of power required by the demand at every hour.

Demand-Supply Requirement

At every time step, the total output of the four generators must exactly match the corresponding demand value.

8.7 Addition of Ramp-Rate Constraints

The ramp-rate constraints are then imposed to restrict the change in generator output between consecutive time steps.

The code uses:

```
for k = 2:N

    optimizer.subject_to(
        abs(x1(k)-x1(k-1)) <= 160)

    optimizer.subject_to(
        abs(x2(k)-x2(k-1)) <= 160)

    optimizer.subject_to(
        abs(x3(k)-x3(k-1)) <= 160)

    optimizer.subject_to(
        abs(x4(k)-x4(k-1)) <= 160)

end
```

Thus, the change in the output of each generator is limited to

$$|x_i(k) - x_i(k-1)| \leq 160 \text{ MW} \quad (110)$$

The initial transition from the known generation at $k = -1$ to the generation at $k = 0$ is also included separately:

```
optimizer.subject_to(abs(x1(1)-x1_init) <= 160)
optimizer.subject_to(abs(x2(1)-x2_init) <= 160)
optimizer.subject_to(abs(x3(1)-x3_init) <= 160)
optimizer.subject_to(abs(x4(1)-x4_init) <= 160)
```

This ensures that the first scheduled operating point is also consistent with the ramp-rate limitation.

8.8 IPOPT Solver

After defining all decision variables, the objective function, and the constraints, the **IPOPT** solver is selected:

```
optimizer.solver('ipopt');
```

IPOPT is used to numerically solve the resulting constrained nonlinear optimization problem.

The solver searches for a feasible combination of generator setpoints that minimizes the total generation cost while simultaneously satisfying:

- ▶ Demand-supply equality
- ▶ Generator minimum and maximum limits
- ▶ Ramp-rate constraints
- ▶ Initial ramp-rate constraints

Solver

The formulated nonlinear optimization problem is solved using the IPOPT solver through the CasADi `Opti` framework.

8.9 Solving the Optimization Problem

Once the complete optimization problem has been formulated, the solver is executed using:

```
sol = optimizer.solve();
```

The obtained solution contains the optimal generator trajectories. These values are extracted using:

```
x1_opti_casadi = sol.value(x1);  
x2_opti_casadi = sol.value(x2);  
x3_opti_casadi = sol.value(x3);  
x4_opti_casadi = sol.value(x4);
```

The variables `x1_opti_casadi`, `x2_opti_casadi`, `x3_opti_casadi`, and `x4_opti_casadi` therefore contain the **optimal generation setpoints** for the four generators from $k = 0$ to $k = 10$.

These optimized values are subsequently used to calculate the total generation and verify the constraints through the plots generated in the MATLAB code.

8.10 Overall Optimization Procedure

The complete methodology can therefore be summarized as follows:

Overall Optimization Workflow



This procedure results in an optimal generation schedule that minimizes the total operating cost while satisfying the required electrical demand and all generator operating constraints.

9 Optimization Problem in Compact Mathematical Form

The complete generator scheduling problem can be represented as a **single constrained nonlinear optimization problem**. This mathematical formulation combines the objective of minimizing the total generation cost with all the operating constraints discussed in the previous sections.

The optimization variables are the generation setpoints of the four generators:

$$x_i(k), \quad i = 1, 2, 3, 4, \quad k = 0, 1, \dots, 10. \quad (111)$$

The complete optimization problem is formulated as

$$\begin{aligned} & \min_{\{x_i(k)\}} \sum_{k=0}^{10} \sum_{i=1}^4 C_i(x_i(k)) \\ \text{subject to } & \sum_{i=1}^4 x_i(k) = d_k, \quad k = 0, \dots, 10, \\ & x_i^{\min} \leq x_i(k) \leq x_i^{\max}, \quad i = 1, \dots, 4, \quad k = 0, \dots, 10, \\ & |x_i(k) - x_i(k-1)| \leq 160 \text{ MW}, \quad i = 1, \dots, 4, \quad k = 1, \dots, 10, \\ & |x_i(0) - x_i(-1)| \leq 160 \text{ MW}, \quad i = 1, \dots, 4. \end{aligned} \quad (112)$$

This formulation represents the complete optimization problem implemented in the MATLAB/CasADi program.

Complete Optimization Problem

The optimization determines the generation setpoints of all four generators over the 11 time points such that the total generation cost is minimized while satisfying the demand, generator operating limits, and ramp-rate constraints.

9.1 Objective Function

The first part of the formulation is the objective:

$$\min_{\{x_i(k)\}} \sum_{k=0}^{10} \sum_{i=1}^4 C_i(x_i(k)) \quad (113)$$

Here, $C_i(x_i(k))$ represents the cost of generating $x_i(k)$ MW from generator i at hour k .

The inner summation

$$\sum_{i=1}^4 C_i(x_i(k)) \quad (114)$$

gives the total generation cost of all four generators at a particular hour. The outer summation,

$$\sum_{k=0}^{10}, \quad (115)$$

then adds the generation costs over the complete planning horizon.

Therefore, the objective function represents the **total generation cost over all four generators and all 11 time points**.

The optimizer searches for the generator setpoints that produce the minimum value of this total cost. The assignment defines the objective in terms of summing the costs of all generators over the entire planning horizon.

9.2 Demand-Supply Constraint

The first constraint is the demand-supply equality:

$$\sum_{i=1}^4 x_i(k) = d_k, \quad k = 0, \dots, 10 \quad (116)$$

Expanding the summation gives

$$x_1(k) + x_2(k) + x_3(k) + x_4(k) = d_k \quad (117)$$

This constraint ensures that the total power generated by the four generators is exactly equal to the electrical demand at every hour.

For example, at $k = 0$,

$$x_1(0) + x_2(0) + x_3(0) + x_4(0) = 2929.83 \text{ MW} \quad (118)$$

At $k = 2$,

$$x_1(2) + x_2(2) + x_3(2) + x_4(2) = 3537.73 \text{ MW} \quad (119)$$

Thus, the optimization cannot choose generator outputs solely based on cost; the combined generation must always satisfy the required demand.

9.3 Generation-Limit Constraints

The second set of constraints defines the allowable operating range of each generator:

$$x_i^{\min} \leq x_i(k) \leq x_i^{\max} \quad (120)$$

For the four generators used in this problem, this becomes

$$100 \leq x_1(k) \leq 950 \quad (121)$$

$$120 \leq x_2(k) \leq 1000 \quad (122)$$

$$\boxed{150 \leq x_3(k) \leq 900} \quad (123)$$

$$\boxed{100 \leq x_4(k) \leq 980} \quad (124)$$

These constraints ensure that the optimized generation values remain within the specified minimum and maximum operating limits of each generator.

Generator Operating Limits

The optimizer is allowed to select generator outputs only within the specified operating ranges. This prevents infeasible generation levels from being selected.

9.4 Ramp-Rate Constraints

The third constraint limits how rapidly each generator can change its output between consecutive hours:

$$\boxed{|x_i(k) - x_i(k-1)| \leq 160 \text{ MW}} \quad (125)$$

This means that the generation output of each generator can increase or decrease by a maximum of **160 MW** between consecutive time steps.

The constraint can equivalently be written as

$$\boxed{-160 \leq x_i(k) - x_i(k-1) \leq 160} \quad (126)$$

This prevents the optimization algorithm from producing a schedule with unrealistically large changes in generator output.

9.5 Initial Ramp-Rate Constraint

In addition to the ramp-rate constraints within the planning horizon, the transition from the known previous operating point $k = -1$ to the first scheduled point $k = 0$ must also satisfy the ramp-rate limit:

$$\boxed{|x_i(0) - x_i(-1)| \leq 160 \text{ MW}} \quad (127)$$

The known values at $k = -1$ are

$$x_1(-1) = 724.4 \text{ MW}, \quad (128)$$

$$x_2(-1) = 865.9 \text{ MW}, \quad (129)$$

$$x_3(-1) = 819.5 \text{ MW}, \quad (130)$$

$$x_4(-1) = 783.7 \text{ MW}. \quad (131)$$

Therefore, the first scheduled generator outputs must be consistent with these initial operating conditions.

Initial Operating Condition

The first optimized generation point at $k = 0$ is not independent of the previous operating point. The transition from $k = -1$ to $k = 0$ must satisfy the 160 MW ramp-rate limit.

The assignment explicitly includes this initial transition as part of the ramp-rate requirement.

9.6 Complete Interpretation

The compact mathematical formulation can therefore be interpreted as follows:

Interpretation of the Optimization Problem

Find the power output of each of the four generators at every hour from $k = 0$ to $k = 10$ such that the total generation cost is minimized, while the total generation exactly matches the demand, every generator remains within its allowable generation limits, and the change in generator output between consecutive hours does not exceed 160 MW.

Thus, the optimization simultaneously considers **economic performance and physical operating constraints**. A solution is considered acceptable only when it satisfies all of these conditions.

The resulting optimal values

$$\boxed{x_1^*(k), \quad x_2^*(k), \quad x_3^*(k), \quad x_4^*(k)} \quad (132)$$

represent the **optimal generation schedule** obtained from the CasADi-IPOPT optimization.

These values are subsequently used to generate the demand-supply, generator-limit, and ramp-rate plots required for verification.

10 Results and Discussion

After formulating the economic scheduling problem and defining the objective function and operating constraints, the optimization problem is solved using the **CasADi Opti framework** with the **IPOPT solver**. The optimization is performed over the complete planning horizon of

$$\boxed{k = 0, 1, \dots, 10} \quad (133)$$

The optimization problem is solved using the MATLAB command:

```
sol = optimizer.solve();
```

Once the optimization is successfully solved, the optimal values of the four generator trajectories are extracted using the `sol.value()` function:

```
x1_opti_casadi = sol.value(x1);
x2_opti_casadi = sol.value(x2);
x3_opti_casadi = sol.value(x3);
x4_opti_casadi = sol.value(x4);
```

The four resulting vectors contain the optimal generation setpoints of generators G_1 , G_2 , G_3 , and G_4 , respectively, for all the time steps from $k = 0$ to $k = 10$.

The obtained results are analyzed in terms of the generator operating trajectories, total generation, demand satisfaction, generation limits, and ramp-rate constraints.

10.1 Optimal Generator Setpoints

The primary result of the optimization is the optimal power output of each generator at every time step. The optimal generator setpoints are represented by

$$\boxed{x_1^*(k), \quad x_2^*(k), \quad x_3^*(k), \quad x_4^*(k)} \quad (134)$$

where the superscript $*$ indicates that the values are obtained from the optimization problem.

The optimal setpoints obtained from the MATLAB/CasADi solution are presented in Table 7.

Table 7: Optimal Generator Setpoints

Hour k	$x_1^*(k)$ (MW)	$x_2^*(k)$ (MW)	$x_3^*(k)$ (MW)	$x_4^*(k)$ (MW)
0	560	700	750	920
1	620	850	900	980
2	658	1000	900	980
3	635	960	900	980
4	635	880	900	980
5	478	730	850	964
6	461	652	850	980
7	348	513	696	886
8	330	504	687	797
9	478	657	843	971
10	565	809	900	979

The table shows that the generator outputs vary throughout the planning horizon according to the changing demand and the operating characteristics of the generators. The

generators do not share the load equally because they have different nonlinear cost functions and different generation limits.

For example, at $k = 2$, when the system demand reaches its highest value of approximately 3537.73 MW, generators 2, 3, and 4 operate at or close to their upper limits, while the remaining required generation is supplied by generator 1. On the other hand, at $k = 8$, when the demand reaches its lowest value of approximately 2318.16 MW, the outputs of all four generators are reduced.

Therefore, the optimal setpoints represent the generation allocation selected by the optimization algorithm to satisfy the changing demand while minimizing the total generation cost.

10.2 Total Generation and Demand

After obtaining the optimal generator outputs, the total generation at each time step is calculated as

$$d_{\text{calc}}(k) = x_1^*(k) + x_2^*(k) + x_3^*(k) + x_4^*(k) \quad (135)$$

In the MATLAB implementation, this is calculated using:

```
d(k) = x1_opti_casadi(k) +  
        x2_opti_casadi(k) +  
        x3_opti_casadi(k) +  
        x4_opti_casadi(k);
```

The calculated total generation is then compared with the original demand data.

The required condition is

$$d_{\text{calc}}(k) = d_k \quad (136)$$

for

$$k = 0, 1, \dots, 10. \quad (137)$$

The first plot shows the total generation and the specified demand over the complete planning horizon. The two curves overlap, indicating that the optimized generator outputs collectively provide the required power at every time step.

The demand initially increases from approximately **2929.83 MW** at $k = 0$ to its maximum value of **3537.73 MW** at $k = 2$. It then decreases until $k = 8$, where the demand reaches its minimum value of approximately **2318.16 MW**. After $k = 8$, the demand increases again and reaches approximately **3253.15 MW** at $k = 10$.

The optimized total generation follows the same variation. Hence, the result confirms that the **demand-supply equality constraint is satisfied throughout the planning horizon**. The assignment specifically requires this comparison to demonstrate that total generation matches the demand at every hour.

10.3 Variation of Generator Outputs

The second set of results shows the variation of the four generator outputs over the planning horizon. The generator trajectories are represented by

$$\boxed{x_1^*(k), \quad x_2^*(k), \quad x_3^*(k), \quad x_4^*(k)} \quad (138)$$

From the plot, it can be observed that the generator outputs change according to the variation in system demand.

At the beginning of the planning horizon, the generator outputs are relatively high because the demand is approximately 2929.83 MW. As the demand increases toward $k = 2$, the generator outputs also increase. At $k = 2$, the total demand reaches approximately 3537.73 MW, resulting in high generation levels from the four units.

After $k = 2$, the demand gradually decreases. Consequently, the generation outputs of the four generators are reduced. The lowest generation levels occur around $k = 8$, corresponding to the minimum demand of approximately 2318.16 MW.

After $k = 8$, the demand begins to increase again, and the generator outputs increase accordingly.

An important observation from the plot is that the generators operate at different output levels. This is due to the different cost functions and operating limits associated with each generator. The optimization algorithm determines the most suitable combination of generator outputs rather than dividing the load equally among the four units.

Thus, the generator trajectories demonstrate how the optimization algorithm coordinates the four generating units to follow the time-varying demand while considering their economic and physical characteristics.

10.4 Verification of Generation Limits

The optimal generator outputs must remain within their specified minimum and maximum generation limits. The limits used in the optimization are

$$\boxed{100 \leq x_1(k) \leq 950} \quad (139)$$

$$\boxed{120 \leq x_2(k) \leq 1000} \quad (140)$$

$$\boxed{150 \leq x_3(k) \leq 900} \quad (141)$$

and

$$\boxed{100 \leq x_4(k) \leq 980} \quad (142)$$

The second figure plots the optimal trajectory of each generator together with its corresponding lower and upper generation limits.

From the figure, the generator trajectories remain within their respective permissible operating ranges throughout the planning horizon. Some generators reach or operate very close to their upper limits during periods of high demand.

For example:

- ▶ Generator 2 reaches its upper limit of approximately 1000 MW around $k = 2$.
- ▶ Generator 3 reaches its upper limit of approximately 900 MW during periods of high demand.
- ▶ Generator 4 operates close to its upper limit during several high-demand periods.
- ▶ Generator 1 remains within its specified range throughout the horizon.

This confirms that the optimized generation schedule respects the specified generation limits. The assignment requires the generator trajectories and their limits to be presented to demonstrate this constraint satisfaction.

10.5 Generator Output Changes and Ramp-Rate Verification

The next step is to verify whether the optimized generator trajectories satisfy the ramp-rate constraint.

For each generator, the change in generation between two consecutive time steps is calculated as

$$\Delta x_i(k) = |x_i^*(k) - x_i^*(k-1)| \quad (143)$$

The maximum allowable change is

$$\Delta x_i(k) \leq 160 \text{ MW} \quad (144)$$

In the MATLAB code, the change in generator 1 output is calculated as:

```
diff1(k) = abs(x1_opti_casadi(k) -  
              x1_opti_casadi(k-1));
```

The same calculation is performed for generators 2, 3, and 4.

The resulting values are plotted to determine whether any generator exceeds the specified 160 MW ramp-rate limit. If all the calculated changes remain below or equal to 160 MW, the ramp-rate constraint is satisfied.

The initial transition must also be considered:

$$|x_i^*(0) - x_i^*(-1)| \leq 160 \text{ MW} \quad (145)$$

This condition ensures that the first optimized setpoint at $k = 0$ can be reached from the known generator operating point at $k = -1$.

The initial values are

$$x_1(-1) = 724.4 \text{ MW}, \quad (146)$$

$$x_2(-1) = 865.9 \text{ MW}, \quad (147)$$

$$x_3(-1) = 819.5 \text{ MW}, \quad (148)$$

$$x_4(-1) = 783.7 \text{ MW}. \quad (149)$$

The assignment specifically requires this initial transition to be included in the ramp-rate verification.

10.6 Overall Feasibility of the Optimal Solution

The optimized solution is considered feasible only when all the constraints of the scheduling problem are satisfied simultaneously.

The first condition is the demand-supply equality:

$$\boxed{x_1^*(k) + x_2^*(k) + x_3^*(k) + x_4^*(k) = d_k} \quad (150)$$

The second condition is that every generator remains within its operating limits:

$$\boxed{x_i^{\min} \leq x_i^*(k) \leq x_i^{\max}} \quad (151)$$

The third condition is the ramp-rate limitation:

$$\boxed{|x_i^*(k) - x_i^*(k-1)| \leq 160 \text{ MW}} \quad (152)$$

Finally, the initial transition must satisfy

$$\boxed{|x_i^*(0) - x_i^*(-1)| \leq 160 \text{ MW}} \quad (153)$$

Based on the obtained plots and the optimization formulation, the generation trajectories satisfy the required demand, remain within the specified generation limits, and are restricted by the specified ramp-rate constraints.

Therefore, the obtained solution represents a **feasible generation schedule** that satisfies the physical operating requirements of the four-generator system while minimizing the defined total generation cost.

Feasibility Condition

The optimal schedule is feasible only when all of the following conditions are satisfied simultaneously:

Total Generation = Demand,
Generator Outputs within Operating Limits,
Ramp Rate ≤ 160 MW,
Initial Transition Ramp Rate ≤ 160 MW.

10.7 Discussion of the Optimization Results

The optimization results demonstrate the ability of the **CasADi–IPOPT framework** to determine an economically efficient generation schedule for a system containing four generators with different nonlinear cost characteristics.

The generator outputs vary throughout the planning horizon in response to the changing electrical demand. When the demand increases, the optimizer increases the total generation by appropriately adjusting the generator setpoints. Similarly, when the demand decreases, the generator outputs are reduced while respecting their minimum generation limits and ramp-rate restrictions.

The generation is not distributed equally among the four generators. Instead, the optimization algorithm determines the generation combination based on the individual cost functions and the operating constraints of each generator. During periods of high demand, some generators operate close to their maximum generation limits, while during periods of lower demand their outputs are reduced.

Three important observations can be made from the results:

1. Demand Tracking:

The sum of the four optimized generator outputs follows the specified demand profile at every time step, confirming satisfaction of the demand-supply equality constraint.

2. Generation-Limit Compliance:

The output of every generator remains within its specified minimum and maximum generation limits. Some generators reach their upper limits during high-demand periods, but do not exceed them.

3. Ramp-Rate Compliance:

The changes in generator output between consecutive time steps are restricted by the 160 MW ramp-rate limit, ensuring that the obtained generation schedule does not require excessively rapid changes in generator output.

Overall Result

The optimization successfully determines a **feasible and economical generation schedule** for the four-generator system. The optimized schedule simultaneously considers the **time-varying electrical demand, nonlinear generation costs, generator operating limits, and ramp-rate limitations**.

Overall, the results show that the optimization successfully determines a feasible and economical generation schedule for the four-generator system. The optimized schedule simultaneously considers the **time-varying electrical demand, nonlinear generation costs, generator operating limits, and ramp-rate limitations**. This demonstrates the effectiveness of the formulated nonlinear optimization approach for economic generator scheduling.

11 Figure 1 — Total Generation vs. Demand

11.1 Figure 12.1: Total Generation and Demand versus Hour Index

The first figure presents a comparison between the **total power generated by the four generators** and the **electrical demand** over the complete planning horizon. The purpose of this plot is to verify whether the optimized generator outputs collectively satisfy the required system demand at every time step.

The total generation at hour k is calculated from the optimal generator setpoints as

$$P_{\text{gen}}(k) = x_1^*(k) + x_2^*(k) + x_3^*(k) + x_4^*(k) \quad (154)$$

where $x_i^*(k)$ represents the optimized power output of generator i at hour k .

According to the demand-supply equality constraint imposed in the optimization problem,

$$x_1^*(k) + x_2^*(k) + x_3^*(k) + x_4^*(k) = d_k \quad (155)$$

for

$$k = 0, 1, \dots, 10. \quad (156)$$

Therefore,

$$P_{\text{gen}}(k) = d_k \quad (157)$$

must hold at every time step.

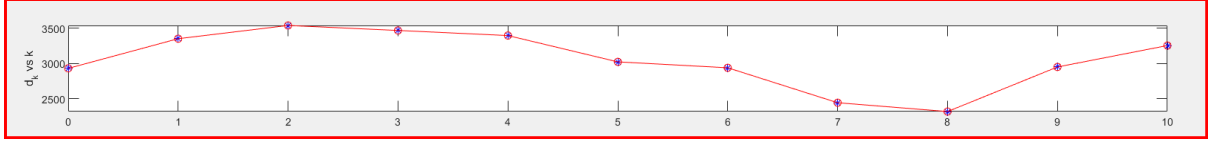


Figure 1: Total generation and electrical demand versus hour index.

As shown in Figure 1, the total generation and demand curves overlap throughout the entire planning horizon. This indicates that the sum of the four optimized generator outputs is equal to the corresponding demand at each hour.

11.2 Observation from the Figure

From the obtained plot, the **total generation curve and the demand curve overlap throughout the entire planning horizon**. This indicates that the sum of the four optimized generator outputs is equal to the corresponding demand at each hour.

At $k = 0$, the required demand is approximately

$$d_0 = 2929.83 \text{ MW} \quad (158)$$

As the demand increases, the total generation also increases. The demand reaches its maximum value at $k = 2$:

$$d_2 = 3537.73 \text{ MW} \quad (159)$$

After $k = 2$, the demand decreases gradually, and the total generation follows the same decreasing trend. The minimum demand occurs at $k = 8$:

$$d_8 = 2318.16 \text{ MW} \quad (160)$$

Following this minimum, the demand increases again, reaching

$$d_{10} = 3253.15 \text{ MW} \quad (161)$$

at the final time step. The total generation follows this increase as well.

The overlapping curves therefore demonstrate that the optimization has successfully allocated the generation among the four units such that the **generation-demand balance is maintained at every time step**.

11.3 Verification of the Equality Constraint

The MATLAB code calculates the total generation using

$$d(k) = x1_opti_casadi(k) + x2_opti_casadi(k) +$$

$x3_opti_casadi(k) +$
 $x4_opti_casadi(k);$

This calculated value is then plotted against the original demand data.

The fact that the two curves coincide provides a graphical verification of

$$d_{\text{calc}}(k) - d_k = 0 \quad (162)$$

for the complete scheduling horizon.

Equality Constraint Verification

The overlapping total-generation and demand curves confirm that

$$P_{\text{gen}}(k) = d_k$$

at every time step $k = 0, \dots, 10$.

Therefore, there is no generation-demand mismatch in the optimized schedule.

Hence, Figure 1 confirms that the **demand-supply equality constraint is satisfied by the optimized solution**. The assignment requires this figure specifically to demonstrate that the optimal total generation matches the demand at every time step.

11.4 Conclusion from Figure 1

The result confirms that the four generators collectively respond to the time-varying load requirement. Even though the individual generator outputs are determined independently by the optimization variables, their combined output exactly follows the specified demand profile at every time step.

Therefore, the obtained generation schedule maintains the required **power balance throughout the 10-hour planning horizon**.

Figure 12.1 – Key Finding

The total optimized generation follows the demand profile exactly. The overlap of the two curves provides a clear graphical verification that the demand-supply equality constraint is satisfied throughout the complete scheduling horizon.

12 Figure 2 — Generator Setpoints

12.1 Figure 13.1: Optimal Generator Setpoints and Operating Limits

The second figure presents the **optimal power generation trajectories of the four generators** over the complete planning horizon. The figure is used to analyze how the optimization algorithm distributes the required electrical demand among the four generators at each time step while respecting the operating limits of each generating unit.

The four generator outputs are represented by

$$\boxed{x_1^*(k), \quad x_2^*(k), \quad x_3^*(k), \quad x_4^*(k)} \quad (163)$$

where $x_i^*(k)$ denotes the optimized power output of generator i at hour k .

The horizontal axis represents the hour index,

$$\boxed{k = 0, 1, \dots, 10} \quad (164)$$

while the vertical axis represents the generator power output in MW.

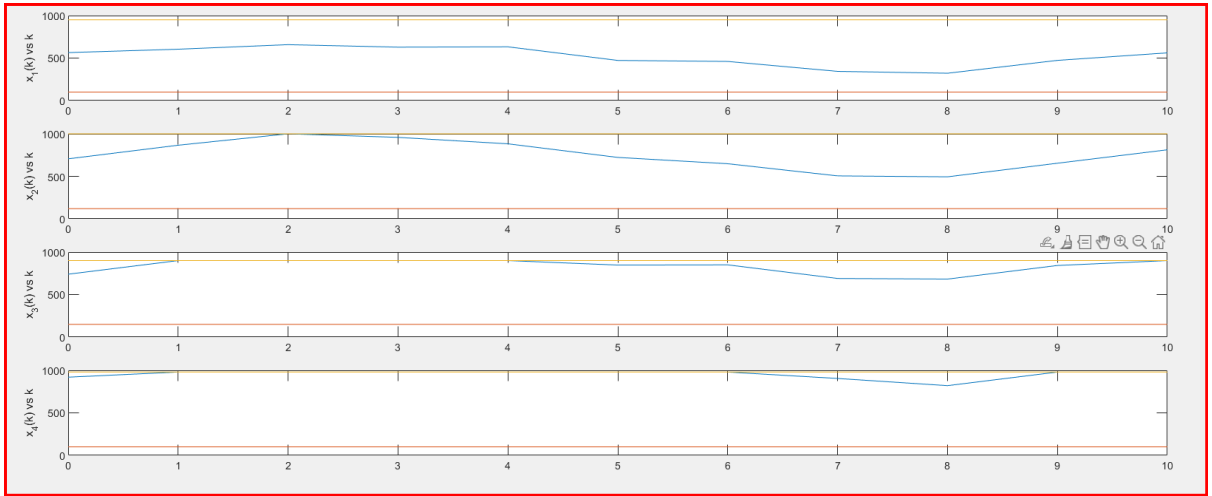


Figure 2: Optimal generator setpoints and their corresponding operating limits over the planning horizon.

12.2 Generator 1

For generator 1, the optimized trajectory is represented by

$$\boxed{x_1^*(k)} \quad (165)$$

The permissible operating range is

$$\boxed{100 \leq x_1(k) \leq 950 \text{ MW}} \quad (166)$$

In Figure 2, the optimal trajectory of generator 1 remains between its lower limit of 100 MW and upper limit of 950 MW throughout the planning horizon.

The generator output initially increases as the demand increases, reaching a relatively high value around $k = 2$. As the system demand decreases after $k = 2$, the output of generator 1 also decreases. The output reaches a lower level around $k = 8$, corresponding to the lower system demand, and then increases again as the demand rises toward $k = 10$.

Thus, generator 1 adjusts its output according to the changing demand while remaining within its allowable operating range.

12.3 Generator 2

For generator 2, the optimized trajectory is represented by

$$\boxed{x_2^*(k)} \quad (167)$$

Its operating limits are

$$\boxed{120 \leq x_2(k) \leq 1000 \text{ MW}} \quad (168)$$

The generator 2 trajectory shows a significant variation over the planning horizon. Its output increases during the initial increase in demand and reaches approximately its upper generation limit during the high-demand period around $k = 2$.

As the demand decreases, the output of generator 2 is also reduced. Around $k = 8$, when the system demand is at its minimum, generator 2 operates at a considerably lower output. When the demand increases again after $k = 8$, its generation output increases correspondingly.

The trajectory remains within the specified range of 120 MW to 1000 MW, demonstrating compliance with the generation-limit constraint.

12.4 Generator 3

The optimized output of generator 3 is represented by

$$\boxed{x_3^*(k)} \quad (169)$$

Its allowable generation range is

$$\boxed{150 \leq x_3(k) \leq 900 \text{ MW}} \quad (170)$$

Generator 3 operates at relatively high generation levels during the high-demand portion of the planning horizon. Around $k = 1$ to $k = 4$, its output reaches or remains close to its upper limit of 900 MW.

As the demand decreases, the output of generator 3 also decreases. Its output reaches a lower level around $k = 7$ – 8 , after which it increases again as the demand rises.

The trajectory remains within the 150 MW to 900 MW operating range, confirming that the optimized solution respects the specified limits of generator 3.

12.5 Generator 4

The optimized trajectory of generator 4 is represented by

$$\boxed{x_4^*(k)} \quad (171)$$

The operating limits for this generator are

$$\boxed{100 \leq x_4(k) \leq 980 \text{ MW}} \quad (172)$$

Generator 4 operates at relatively high power levels for most of the planning horizon. During the high-demand periods, its output approaches the upper limit of 980 MW.

As the demand decreases toward $k = 8$, the output of generator 4 also decreases. When the demand increases again after $k = 8$, its output rises accordingly.

The generator remains within its specified operating range throughout the scheduling horizon.

12.6 Overall Observation from Figure 2

Figure 2 demonstrates that the four generators respond to the variation in system demand by changing their individual power outputs. However, the changes are not identical for all four generators because each unit has a different nonlinear generation cost function and different operating limits.

During periods of **high demand**, such as around $k = 1$ – 4 , several generators operate at relatively high output levels, with some approaching their upper generation limits. During periods of **low demand**, particularly around $k = 7$ – 8 , the generator outputs are reduced.

The optimized trajectories can therefore be interpreted as the result of the economic scheduling algorithm balancing two requirements:

$$\boxed{\text{Satisfy the required demand}} \quad (173)$$

while simultaneously

$$\boxed{\text{Minimize the total generation cost}} \quad (174)$$

At the same time, the generation outputs must remain within their respective operating limits.

The four operating ranges used in the optimization are shown in Table 8.

Table 8: Operating limits of the four generators

Generator	Minimum Limit (MW)	Maximum Limit (MW)
G_1	100	950
G_2	120	1000
G_3	150	900
G_4	100	980

Therefore, Figure 2 provides a graphical verification that the **optimal generator set-points remain within their permitted operating ranges** throughout the planning horizon.

This directly satisfies the requirement for the second figure of the assignment.

12.7 Conclusion from Figure 2

Overall, Figure 2 shows that the optimization algorithm successfully determines the individual generator setpoints for the varying load conditions.

The generators operate at different power levels based on their cost characteristics and constraints, while their outputs remain within the specified minimum and maximum limits.

Key Finding

The generator trajectories remain within their respective operating limits throughout the scheduling horizon. The results therefore confirm that the obtained generation schedule is **operationally feasible with respect to the generator capacity constraints**.

Thus, Figure 2 demonstrates that the four generators are appropriately coordinated by the optimization algorithm to respond to the varying electrical demand while respecting their individual generation limits.

13 Figure 3 — Ramp-Rate Changes

13.1 Figure 14.1: Ramp-Rate Changes of the Four Generators

The third figure is used to verify whether the **ramp-rate constraints** imposed on the four generators are satisfied by the optimized generation schedule. The ramp-rate constraint limits the amount by which a generator can increase or decrease its power output between two consecutive time steps.

For each generator i , the change in power output between two consecutive hours is calculated as

$$\boxed{\Delta x_i(k) = |x_i^*(k) - x_i^*(k-1)|} \quad (175)$$

where $x_i^*(k)$ is the optimized power output of generator i at time step k .

The maximum allowable change specified in the problem is

$$\boxed{\Delta x_i(k) \leq 160 \text{ MW}} \quad (176)$$

Therefore, the ramp-rate plot shows the values of

$$|x_1^*(k) - x_1^*(k-1)|, \quad (177)$$

$$|x_2^*(k) - x_2^*(k-1)|, \quad (178)$$

$$|x_3^*(k) - x_3^*(k-1)|, \quad (179)$$

and

$$|x_4^*(k) - x_4^*(k-1)| \quad (180)$$

for the consecutive time steps.

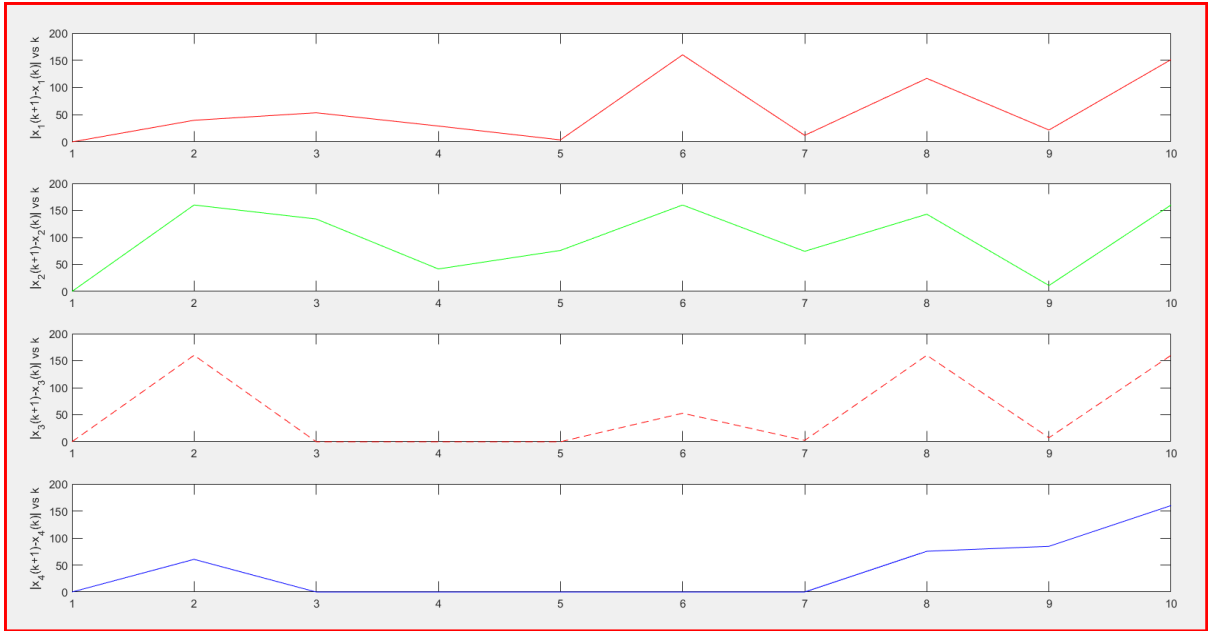


Figure 3: Ramp-rate changes of the four generators over the scheduling horizon.

13.2 Calculation of Ramp-Rate Changes

After obtaining the optimal generator trajectories from CasADi, the MATLAB code calculates the change in output between consecutive time steps.

For generator 1, for example,

$$\text{diff1}(k) = \text{abs}(x1_opti_casadi(k) - x1_opti_casadi(k-1));$$

Similarly, the corresponding changes for generators 2, 3, and 4 are calculated as

$$\text{diff2}(k) = \text{abs}(x2_opti_casadi(k) - x2_opti_casadi(k-1));$$

$$\text{diff3}(k) = \text{abs}(x3_opti_casadi(k) - x3_opti_casadi(k-1));$$

$$\text{diff4}(k) = \text{abs}(x4_opti_casadi(k) - x4_opti_casadi(k-1));$$

These calculated values represent the magnitude of the change in generator output from one time step to the next.

Since the absolute value is used, both an increase and a decrease in generation are treated in the same way. For example, if a generator changes from 700 MW to 820 MW,

$$\Delta x = |820 - 700| = 120 \text{ MW}. \quad (181)$$

Similarly, if its output changes from 820 MW to 700 MW,

$$\Delta x = |700 - 820| = 120 \text{ MW}. \quad (182)$$

In both cases, the magnitude of the generation change is 120 MW.

13.3 Ramp-Rate Limit Verification

The maximum permissible change is

$$\boxed{160 \text{ MW}} \quad (183)$$

Therefore, every value shown in the ramp-rate plot must satisfy

$$\boxed{\Delta x_i(k) \leq 160 \text{ MW}} \quad (184)$$

If a calculated value is less than or equal to 160 MW, the corresponding transition satisfies the ramp-rate constraint. If a value were greater than 160 MW, that transition would violate the constraint and the corresponding generation schedule would not be feasible.

Thus, the ramp-rate plot provides a direct graphical method of checking the feasibility of the optimized generator trajectories.

The assignment requires the ramp-rate changes of all four generators to be plotted and compared with the specified **160 MW limit**.

13.4 Initial Transition from $k = -1$ to $k = 0$

A special consideration is required for the first time step. The ramp-rate calculation does not begin only between $k = 0$ and $k = 1$. The transition from the known operating point at $k = -1$ to the first optimized operating point at $k = 0$ must also satisfy the ramp-rate constraint.

The initial condition is given by

$$x_1(-1) = 724.4 \text{ MW}, \quad (185)$$

$$x_2(-1) = 865.9 \text{ MW}, \quad (186)$$

$$x_3(-1) = 819.5 \text{ MW}, \quad (187)$$

$$x_4(-1) = 783.7 \text{ MW}. \quad (188)$$

Therefore, the initial changes are

$$\boxed{|x_1^*(0) - 724.4| \leq 160} \quad (189)$$

$$\boxed{|x_2^*(0) - 865.9| \leq 160} \quad (190)$$

$$\boxed{|x_3^*(0) - 819.5| \leq 160} \quad (191)$$

and

$$\boxed{|x_4^*(0) - 783.7| \leq 160} \quad (192)$$

These conditions ensure that the optimized generation schedule can be reached from the actual operating condition immediately before the scheduling horizon begins.

The assignment explicitly requires the transition from $k = -1$ to $k = 0$ to satisfy the same 160 MW ramp-rate limit.

13.5 Interpretation of the Ramp-Rate Plot

The ramp-rate plot allows the generation changes of all four generators to be compared over the entire planning horizon.

When the demand increases, the generators must increase their outputs to provide the additional required power. When the demand decreases, their outputs must decrease accordingly. However, the optimization cannot make these changes arbitrarily large because of the 160 MW ramp-rate restriction.

Therefore, the optimized generation trajectories shown in Figure 2 and the ramp-rate values shown in Figure 3 are directly related. The changes in the generator trajectories are restricted by the ramp-rate constraints.

For each generator, the calculated ramp-rate values should remain at or below the horizontal limit of

$$\boxed{160 \text{ MW}} \quad (193)$$

This demonstrates that the generator schedule is not only economically optimized but also respects the allowable rate of change of generator output.

Ramp-Rate Verification

The ramp-rate constraint requires

$$\boxed{|x_i^*(k) - x_i^*(k-1)| \leq 160 \text{ MW}}$$

for every generator and every applicable consecutive time step.

13.6 Conclusion from Figure 3

Figure 3 provides the final constraint verification for the generator scheduling problem. By calculating

$$\boxed{|x_i^*(k) - x_i^*(k-1)|} \quad (194)$$

for all four generators and comparing these values with the 160 MW limit, the feasibility of the generator transitions can be verified.

From the obtained figure, the ramp-rate changes of the four generators remain within the specified limit of 160 MW. The maximum observed changes are approximately 160 MW or less, with generator 1, generator 2, and generator 3 reaching values close to the limit at some time steps, while generator 4 also remains below the specified limit.

Therefore,

$$\boxed{|x_i^*(k) - x_i^*(k-1)| \leq 160 \text{ MW}} \quad (195)$$

is satisfied for the complete scheduling horizon.

The initial transition from $k = -1$ to $k = 0$ is also constrained by the same 160 MW limit through the initial ramp-rate constraints defined in the optimization problem.

Key Finding

Figure 3 confirms that the optimized generator schedule respects the **ramp-rate limitations**. The changes in generator outputs between consecutive time steps remain within the specified 160 MW operating constraint.

Thus, Figure 3 confirms that the optimized generator schedule respects the **ramp-rate limitations**, ensuring that the changes in generator outputs remain within the specified operating capability.

This completes the verification of the major constraints of the economic scheduling problem.

14 Constraint Verification

The optimization problem contains several constraints that must be satisfied simultaneously for the obtained generator schedule to be considered feasible. Although the corresponding plots provide a visual representation of the results, a separate constraint verification is performed to explicitly confirm that the optimized solution satisfies the mathematical requirements of the problem.

The three main constraints verified are:

1. Demand-supply equality
2. Generator operating limits
3. Ramp-rate limitations

14.1 Demand Constraint Verification

The first constraint to be verified is the **demand-supply equality constraint**. For every hour k , the total power generated by the four generators must be equal to the specified electrical demand.

The condition is

$$x_1^*(k) + x_2^*(k) + x_3^*(k) + x_4^*(k) = d_k \quad (196)$$

or, equivalently, the generation-demand error can be defined as

$$e_d(k) = x_1^*(k) + x_2^*(k) + x_3^*(k) + x_4^*(k) - d_k \quad (197)$$

For a feasible solution,

$$e_d(k) = 0 \quad (198)$$

for

$$k = 0, 1, \dots, 10. \quad (199)$$

In the MATLAB implementation, the total generation is calculated as

$$\begin{aligned} d(k) = & \text{x1_opti_casadi}(k) + \\ & \text{x2_opti_casadi}(k) + \\ & \text{x3_opti_casadi}(k) + \\ & \text{x4_opti_casadi}(k); \end{aligned}$$

This calculated total generation is compared with the original demand data.

From Figure 1, the total generation curve overlaps with the demand curve at all time steps. This confirms that the optimized generator outputs collectively satisfy the required load throughout the planning horizon.

Therefore, the **demand-supply equality constraint is satisfied**.

Demand Constraint Result:

$$x_1^*(k) + x_2^*(k) + x_3^*(k) + x_4^*(k) = d_k$$

The generation-demand balance is satisfied for all $k = 0, \dots, 10$.

14.2 Generation Limit Verification

The second verification checks whether the optimized output of each generator remains within its specified minimum and maximum operating limits.

For each generator, the general condition is

$$x_i^{\min} \leq x_i^*(k) \leq x_i^{\max} \quad (200)$$

for

$$i = 1, 2, 3, 4 \quad (201)$$

and

$$k = 0, 1, \dots, 10. \quad (202)$$

The limits used in the optimization are:

$$100 \leq x_1^*(k) \leq 950 \text{ MW} \quad (203)$$

$$120 \leq x_2^*(k) \leq 1000 \text{ MW} \quad (204)$$

$$\boxed{150 \leq x_3^*(k) \leq 900 \text{ MW}} \quad (205)$$

$$\boxed{100 \leq x_4^*(k) \leq 980 \text{ MW}} \quad (206)$$

The optimal generator trajectories are plotted together with their corresponding lower and upper limits in Figure 2.

From the obtained results, the generator trajectories remain within their respective operating ranges. Some generators reach or approach their upper limits during periods of high demand, but their outputs remain within the specified bounds.

Thus, the **generation-limit constraints are satisfied** for all four generators over the complete planning horizon.

Table 9: Verification of generator operating limits

Generator	Minimum (MW)	Maximum (MW)	Verification
G_1	100	950	Satisfied
G_2	120	1000	Satisfied
G_3	150	900	Satisfied
G_4	100	980	Satisfied

14.3 Ramp-Rate Verification

The third verification concerns the **ramp-rate constraint**, which limits the maximum change in generator output between consecutive time steps.

For each generator, the change in output is calculated as

$$\boxed{\Delta x_i(k) = |x_i^*(k) - x_i^*(k-1)|} \quad (207)$$

The required condition is

$$\boxed{\Delta x_i(k) \leq 160 \text{ MW}} \quad (208)$$

Equivalently,

$$\boxed{|x_i^*(k) - x_i^*(k-1)| \leq 160 \text{ MW}} \quad (209)$$

The MATLAB code calculates these changes for each generator and stores them in `diff1`, `diff2`, `diff3`, and `diff4`. These values are then plotted in Figure 3.

The ramp-rate constraint is satisfied if every calculated change remains less than or equal to 160 MW.

Ramp-Rate Requirement

For every generator and every applicable consecutive time step,

$$|x_i^*(k) - x_i^*(k-1)| \leq 160 \text{ MW}$$

must be satisfied.

14.3.1 Initial Transition Verification

The transition from the known operating point at $k = -1$ to the first optimized point at $k = 0$ is also verified:

$$|x_i^*(0) - x_i(-1)| \leq 160 \text{ MW} \quad (210)$$

The initial generator outputs are

$$x_1(-1) = 724.4 \text{ MW}, \quad (211)$$

$$x_2(-1) = 865.9 \text{ MW}, \quad (212)$$

$$x_3(-1) = 819.5 \text{ MW}, \quad (213)$$

$$x_4(-1) = 783.7 \text{ MW}. \quad (214)$$

Therefore, the initial transition for each generator must also remain within the 160 MW limit.

The corresponding initial constraints are

$$|x_1^*(0) - 724.4| \leq 160 \text{ MW}, \quad (215)$$

$$|x_2^*(0) - 865.9| \leq 160 \text{ MW}, \quad (216)$$

$$|x_3^*(0) - 819.5| \leq 160 \text{ MW}, \quad (217)$$

$$|x_4^*(0) - 783.7| \leq 160 \text{ MW}. \quad (218)$$

These conditions ensure that the first optimized operating point is reachable from the known operating condition before the scheduling horizon begins.

14.4 Overall Constraint Verification

The feasibility of the optimized solution can therefore be summarized using the following conditions.

Table 10: Summary of constraint verification

Constraint	Verification Condition	Result
Demand-supply balance	$x_1^*(k) + x_2^*(k) + x_3^*(k) + x_4^*(k) = d_k$	Satisfied
G_1 generation limit	$100 \leq x_1^*(k) \leq 950$ MW	Satisfied
G_2 generation limit	$120 \leq x_2^*(k) \leq 1000$ MW	Satisfied
G_3 generation limit	$150 \leq x_3^*(k) \leq 900$ MW	Satisfied
G_4 generation limit	$100 \leq x_4^*(k) \leq 980$ MW	Satisfied
Ramp-rate constraint	$ x_i^*(k) - x_i^*(k-1) \leq 160$ MW	Satisfied
Initial ramp constraint	$ x_i^*(0) - x_i^*(-1) \leq 160$ MW	Satisfied

Hence, the obtained optimal generation schedule satisfies the **demand requirement, generator operating limits, and ramp-rate limitations**.

This confirms that the solution obtained using the **CasADi-IPOPT optimization framework** is a **feasible operating schedule** for the four-generator system.

14.5 Final Verification Statement

The constraint verification confirms that the optimized generation schedule satisfies all the mathematical and physical requirements defined in the optimization problem.

Specifically,

Demand-supply balance is satisfied,
 All generator outputs remain within their limits,
 All consecutive generation changes remain within 160 MW,
 The initial transition also satisfies the ramp-rate limit.

(219)

Overall Constraint Verification Result

The optimized generation schedule is **feasible** with respect to the demand balance, generator operating limits, and ramp-rate constraints. Therefore, the solution obtained from the CasADi-IPOPT optimization satisfies the required operating conditions of the four-generator economic scheduling problem.

Thus, the constraint verification demonstrates that the optimization does not merely

minimize the generation cost, but produces a **physically feasible and economically optimized generation schedule** for the complete planning horizon.

15 Discussion of Optimal Scheduling

The optimization results provide more than just the numerical values of the generator set-points. They show how the optimization algorithm distributes the electrical load among the four generators while considering their **different generation costs, operating limits, and ramp-rate constraints**.

The purpose of this section is to interpret the obtained generation schedules and explain the behavior of the generators over the planning horizon.

15.1 Generator Power Allocation

From the obtained optimal generation trajectories, it can be observed that the four generators do not contribute equally to the total system generation.

Generator 4 generally operates at a relatively high power level throughout much of the planning horizon. Generator 3 also operates at a high output, particularly during periods of high demand. Generator 2 increases its output significantly during high-demand periods and reaches its upper limit of approximately 1000 MW around the maximum-demand region.

Generator 1 generally operates at a lower output compared with generators 2, 3, and 4. However, its output is adjusted as necessary to satisfy the remaining system demand while maintaining all the imposed constraints.

Therefore, the optimization does not simply divide the load equally among the four generators. Instead, it determines the combination of generator outputs that gives the **minimum total cost** while satisfying the system requirements.

Key Observation

The generation is distributed unequally among the four generators. This unequal allocation results from the interaction between the different nonlinear cost functions and the operating constraints of the generating units.

15.2 Generators Operating Near Their Limits

The generator-limit plots show that some generators operate close to their upper operating limits, particularly during periods of high demand.

The upper limits are

$$\boxed{x_1^{\max} = 950 \text{ MW}} \quad (220)$$

$$\boxed{x_2^{\max} = 1000 \text{ MW}} \quad (221)$$

$$\boxed{x_3^{\max} = 900 \text{ MW}} \quad (222)$$

and

$$\boxed{x_4^{\max} = 980 \text{ MW}} \quad (223)$$

From the obtained trajectories, generators 2, 3, and 4 operate close to their upper limits during the high-demand portion of the scheduling horizon.

For example, around $k = 2$, the demand reaches approximately

$$\boxed{d_2 = 3537.73 \text{ MW}} \quad (224)$$

and the outputs of several generators are close to their maximum allowable values.

This indicates that the available generation capacity of these units is being utilized significantly when the system demand is high. However, the optimization does not allow any generator to exceed its specified maximum limit.

15.3 Redistribution of Generation When Demand Increases

When the electrical demand increases, the total generation must also increase because of the equality constraint

$$\boxed{x_1(k) + x_2(k) + x_3(k) + x_4(k) = d_k} \quad (225)$$

For example, the demand increases from approximately

$$2929.83 \text{ MW} \quad \text{at } k = 0 \quad (226)$$

to

$$3537.73 \text{ MW} \quad \text{at } k = 2. \quad (227)$$

Consequently, the generation outputs of the four generators increase to provide the additional required power.

The optimizer determines how this additional generation should be distributed among the generators. It considers the cost functions and the available generation capacity of each unit rather than increasing every generator by the same amount.

Thus, the increase in demand results in a coordinated redistribution of generation among the available generating units.

High-Demand Operation

As the demand increases, additional generation is allocated among the available units while respecting their individual generation limits and ramp-rate constraints.

15.4 Reduction of Generation When Demand Decreases

The same principle applies when the demand decreases.

After reaching its maximum around $k = 2$, the demand decreases progressively and reaches its minimum value of approximately

$$d_8 = 2318.16 \text{ MW} \quad (228)$$

at $k = 8$.

The generator outputs consequently decrease during this period. The optimized schedule reduces the generation levels while ensuring that no generator goes below its minimum operating limit.

For example, around $k = 7-8$, the outputs of all four generators are considerably lower than their values during the high-demand period. After $k = 8$, the demand begins to increase again, and the generator outputs increase accordingly.

This demonstrates that the optimization responds to both increases and decreases in system demand.

15.5 Effect of Ramp-Rate Constraints

The ramp-rate constraint has an important effect on the shape of the generator trajectories.

The optimization imposes

$$|x_i(k) - x_i(k-1)| \leq 160 \text{ MW} \quad (229)$$

Therefore, even if a large change in generator output could reduce the generation cost, the optimizer cannot make that change if it exceeds the allowable ramp rate.

As a result, the generator trajectories change progressively rather than showing arbitrarily large jumps between consecutive time steps.

The same constraint is also applied to the transition from the known operating condition at $k = -1$ to the first optimized point at $k = 0$:

$$|x_i(0) - x_i(-1)| \leq 160 \text{ MW} \quad (230)$$

Thus, the ramp-rate constraint ensures that the optimal schedule is not only economically desirable but also consistent with the permitted rate of change of generator output.

15.6 Influence of the Nonlinear Cost Functions

Another important factor affecting the generation allocation is the different nonlinear cost function associated with each generator.

The four generators have different cost expressions:

$$C_1(x_1) = 100 + 2.5x_1 + 0.0055x_1^2 + 50 \sin\left(\frac{x_1}{120}\right) \quad (231)$$

$$C_2(x_2) = 120 + 2.2x_2 + 0.0038x_2^2 + 40 \sin\left(\frac{x_2}{140} + 0.5\right) \quad (232)$$

$$C_3(x_3) = 200 + 2.8x_3 + 0.0022x_3^2 + 60 \sin\left(\frac{x_3}{110} - 0.3\right) \quad (233)$$

$$C_4(x_4) = 400 + 3.2x_4 + 0.0015x_4^2 + 45 \sin\left(\frac{x_4}{130} + 1.0\right) \quad (234)$$

Because these functions are different, the cost associated with producing an additional MW is not the same for all generators and does not vary in the same manner with generation level.

The sinusoidal terms also introduce nonlinear variations in the generation costs. Consequently, the optimizer must evaluate the combined effect of all four cost functions when determining the optimal generation allocation.

Therefore, the resulting generator outputs are a consequence of the interaction between:

- ▶ The nonlinear generation costs,
- ▶ Generator minimum and maximum limits,
- ▶ Ramp-rate limitations, and
- ▶ The time-varying electrical demand.

15.7 Overall Interpretation of the Optimal Schedule

The optimal scheduling results can be understood as a continuous balance between **economic optimization and physical feasibility**.

When demand is high, the optimizer utilizes the available generation capacity of the generators while respecting their maximum limits. When demand decreases, generation is reduced while respecting minimum limits and ramp-rate restrictions.

The generators therefore work together to maintain

$$x_1^*(k) + x_2^*(k) + x_3^*(k) + x_4^*(k) = d_k \quad (235)$$

at every time step.

At the same time, the optimizer attempts to minimize

$$\sum_{k=0}^{10} \sum_{i=1}^4 C_i(x_i(k)) \quad (236)$$

Hence, the final schedule is not simply a demand-following schedule. It is an **economically optimized demand-following schedule subject to generator operating constraints**.

15.8 Key Observations

The main observations from the optimal scheduling results are:

1. **Unequal generation allocation:**

The four generators operate at different power levels because their cost functions and operating limits are different.

2. **High utilization during peak demand:**

During the high-demand period around $k = 1-4$, several generators operate close to their upper generation limits.

3. **Reduced generation during low demand:**

As the demand decreases toward $k = 8$, the outputs of the generators are reduced accordingly.

4. **Demand-following behavior:**

When the demand increases after $k = 8$, the generator outputs increase again to satisfy the additional load.

5. **Ramp-rate limitation:**

The 160 MW ramp-rate constraint prevents sudden changes in generator output and produces smoother generation trajectories.

6. **Nonlinear cost influence:**

The different nonlinear cost functions determine how the required load is economically distributed among the four generators.

7. **Feasible operation:**

The final schedule satisfies the demand balance, generator operating limits, and ramp-rate constraints.

Overall Scheduling Result

The results demonstrate that the **CasADi-IPOPT optimization approach** successfully coordinates the four generators to obtain a feasible and economically optimized generation schedule.

The obtained solution balances the changing electrical demand with the different economic and physical characteristics of the generating units while satisfying the specified operating constraints.

Overall, the results demonstrate that the **CasADi-IPOPT optimization approach successfully coordinates the four generators** to obtain a feasible and economically optimized generation schedule. The obtained solution balances the changing electrical demand with the different economic and physical characteristics of the generating units.

16 Conclusion

The economic scheduling problem for a system consisting of **four electrical generators** was successfully formulated and solved as a constrained nonlinear optimization problem. The main objective of the work was to determine the optimal generation setpoints of the four generators over a **10-hour planning horizon** while minimizing the total generation cost and satisfying the required operating constraints.

A nonlinear generation-cost model was considered for each generator. The total objective function was formed by summing the individual generation costs of all four generators over the complete planning horizon. Because the cost functions contain quadratic and sinusoidal terms, the resulting optimization problem is nonlinear.

The optimization was formulated using the **CasADi Opti** framework and solved using the **IPOPT** nonlinear programming solver. The optimizer determined the generation schedule for

$$\boxed{x_1(k), \quad x_2(k), \quad x_3(k), \quad x_4(k)} \quad (237)$$

for

$$\boxed{k = 0, 1, \dots, 10.} \quad (238)$$

Several constraints were incorporated into the optimization to ensure that the resulting schedule was physically feasible.

First, the **demand-supply equality constraint** was enforced so that the total generation from the four generators exactly matched the electrical demand at every time step:

$$\boxed{x_1(k) + x_2(k) + x_3(k) + x_4(k) = d_k} \quad (239)$$

The results showed that the total generation follows the demand profile throughout the planning horizon.

Second, the **minimum and maximum generation limits** of each generator were enforced. These constraints ensured that none of the generators operated outside their specified permissible operating range.

Third, the **160 MW ramp-rate constraint** was imposed to restrict the change in generator output between consecutive time steps. The initial transition from the known operating condition at $k = -1$ to the first optimized point at $k = 0$ was also included in the ramp-rate constraint. This ensured that the obtained generation schedule did not require unrealistically rapid changes in generator output.

The results obtained from the optimization demonstrate that the generator outputs are adjusted according to the variation in electrical demand. During high-demand periods, the generators increase their outputs, with some units operating close to their upper generation limits. During periods of lower demand, the generation outputs are reduced while still respecting the minimum generation limits and ramp-rate restrictions.

Key Result:

The optimized generation schedule follows the time-varying demand while satisfying the generator operating limits and the specified 160 MW ramp-rate constraint.

The constraint-verification results further demonstrate that the optimized schedule satisfies the required **power-balance, generation-limit, and ramp-rate constraints**. Therefore, the obtained solution is not only cost-optimized but also operationally feasible.

The main outcomes of the optimization can be summarized as follows:

1. The total generation cost was minimized over the complete 10-hour planning horizon.
2. The total generation from the four generators was matched to the electrical demand at every time step.
3. All generator outputs remained within their specified minimum and maximum operating limits.
4. The change in generator output between consecutive time steps remained within the specified 160 MW ramp-rate limit.
5. The transition from the initial operating condition at $k = -1$ to the first optimized operating point at $k = 0$ also satisfied the ramp-rate constraint.
6. The different nonlinear cost functions resulted in an unequal and economically optimized allocation of generation among the four generators.

Overall, the work demonstrates the application of **CasADi and IPOPT for nonlinear economic generator scheduling**. The optimization provides an economically optimal

generation schedule over the specified planning horizon while simultaneously satisfying the electrical demand and the physical operating limitations of the four generators.

Final Conclusion

The results confirm the effectiveness of the **CasADi-IPOPT optimization approach** for solving the considered four-generator economic scheduling problem. The obtained solution successfully balances the changing electrical demand with the different economic characteristics and physical operating constraints of the generating units.

Thus, the complete optimization study demonstrates that a **constrained nonlinear optimization approach** can be effectively used to obtain a **feasible and economically optimized generation schedule** for a multi-generator power system.

A Appendix A: MATLAB/CasADi Implementation

The complete MATLAB implementation used to formulate and solve the four-generator economic scheduling problem is provided in this appendix. The code uses the **CasADi Opti** framework to formulate the nonlinear optimization problem and the **IPOPT** solver to obtain the optimal generator schedule.

The program performs the following main operations:

1. Reads the electrical demand data from the Excel file `demand.xlsx`.
2. Defines the planning horizon and initial generator outputs.
3. Creates the CasADi optimization environment.
4. Defines the four generator decision variables.
5. Specifies the minimum and maximum generation limits.
6. Defines the nonlinear cost functions of the four generators.
7. Forms the total generation-cost objective function.
8. Imposes the demand-supply equality constraint.
9. Imposes generator operating limits.
10. Imposes the 160 MW ramp-rate constraints.
11. Includes the initial ramp-rate constraints.
12. Solves the optimization problem using IPOPT.
13. Extracts the optimal generator setpoints.

14. Calculates the generation changes between consecutive time steps.
15. Calculates the total generation.
16. Generates plots for demand tracking, generator setpoints, operating limits, and ramp-rate changes.

A.1 Complete MATLAB Code

The complete MATLAB/CasADi implementation used for the optimization is given below.

Listing 1: Complete MATLAB/CasADi implementation for four-generator economic scheduling

```
clear
close all
clc

import casadi.*

T = readtable('demand.xlsx');
demand = T.demand_MW;

N = 10;
t = [0:1:N]';

% figure
% plot(t,demand,'ro-');

x1_init = 724.4;
x2_init = 865.9;
x3_init = 819.5;
x4_init = 783.7;

%%
% Beginning of CasADi framework

optimizer = casadi.Opti();
optimizer.solver('ipopt');

x1 = optimizer.variable(1,N+1);
x2 = optimizer.variable(1,N+1);
x3 = optimizer.variable(1,N+1);
x4 = optimizer.variable(1,N+1);

% Lower and upper bounds on setpoints
% of the 4 generators

lb1 = 100;  ub1 = 950;
```

```

lb2 = 120;   ub2 = 1000;
lb3 = 150;   ub3 = 900;
lb4 = 100;   ub4 = 980;

```

```

% Creating reference for the plot

```

```

ref11 = ones(size(t))*lb1;
ref12 = ones(size(t))*ub1;

```

```

ref21 = ones(size(t))*lb2;
ref22 = ones(size(t))*ub2;

```

```

ref31 = ones(size(t))*lb3;
ref32 = ones(size(t))*ub3;

```

```

ref41 = ones(size(t))*lb4;
ref42 = ones(size(t))*ub4;

```

```

% x1 = [lb1:10:ub1]';
% x2 = [lb2:10:ub2]';
% x3 = [lb3:10:ub3]';
% x4 = [lb4:10:ub4]';

```

```

c1 = 0;
c2 = 0;
c3 = 0;
c4 = 0;

```

```

% Individual cost functions of the 4 generators

```

```

for k = 1:N+1

```

```

    c1 = c1 + 100 + 2.5*x1(k) + 0.0055*x1(k)^2 + 50*sin(x1(k)/120);

```

```

    c2 = c2 + 120 + 2.2*x2(k) + 0.0038*x2(k)^2 + 40*sin(x2(k)/140 + 0.5);

```

```

    c3 = c3 + 200 + 2.8*x3(k) + 0.0022*x3(k)^2 + 60*sin(x3(k)/110 - 0.3);

```

```

    c4 = c4 + 400 + 3.2*x4(k) + 0.0015*x4(k)^2 + 45*sin(x4(k)/130 + 1.0);

```

```

end

```

```

% Posing the minimization problem

```

```

% optimizer.minimize(c1);
% optimizer.minimize(c2);
% optimizer.minimize(c3);
% optimizer.minimize(c4);

```



```

optimizer.minimize(c1+c2+c3+c4);

% Equality constraints

for k = 1:N+1

    optimizer.subject_to(
        x1(k)+x2(k)+x3(k)+x4(k)==demand(k))

end

% Inequality constraints

optimizer.subject_to(lb1<=x1<=ub1);
optimizer.subject_to(lb2<=x2<=ub2);
optimizer.subject_to(lb3<=x3<=ub3);
optimizer.subject_to(lb4<=x4<=ub4);

% Ramp-rate constraints

for k = 2:N

    optimizer.subject_to(abs(x1(k)-x1(k-1))<=160)

    optimizer.subject_to(abs(x2(k)-x2(k-1))<=160)

    optimizer.subject_to(abs(x3(k)-x3(k-1))<=160)

    optimizer.subject_to(abs(x4(k)-x4(k-1))<=160)

end

% Initial ramp-rate constraints

optimizer.subject_to(
    abs(x1(1)-x1_init)<=160)

optimizer.subject_to(
    abs(x2(1)-x2_init)<=160)

optimizer.subject_to(
    abs(x3(1)-x3_init)<=160)

optimizer.subject_to(
    abs(x4(1)-x4_init)<=160)

% Solve the optimization problem

```

```

sol = optimizer.solve();

% Extract optimal generator outputs

x1_opti_casadi = sol.value(x1)
x2_opti_casadi = sol.value(x2)
x3_opti_casadi = sol.value(x3)
x4_opti_casadi = sol.value(x4)

% Calculate changes between consecutive time steps

for k = 2:N

    diff1(k) = abs(
        x1_opti_casadi(k)-x1_opti_casadi(k-1));

    diff2(k) = abs(
        x2_opti_casadi(k)-x2_opti_casadi(k-1));

    diff3(k) = abs(
        x3_opti_casadi(k)-x3_opti_casadi(k-1));

    diff4(k) = abs(
        x4_opti_casadi(k)-x4_opti_casadi(k-1));

end

% Calculate total generation

for k = 1:N+1

    d(k) =
        x1_opti_casadi(k)
        + x2_opti_casadi(k)
        + x3_opti_casadi(k)
        + x4_opti_casadi(k);

end

d;

% Plot the results

figure

subplot(511)

plot(t(1:end),d', 'b*')

```

```

hold on
plot(t,demand,'ro-');
ylabel('d_k vs k')

subplot(512)

plot(t,x1_opti_casadi)
ylabel('x_1(k) vs k')
hold on
plot(t,ref11)
plot(t,ref12)

subplot(513)

plot(t,x2_opti_casadi)
ylabel('x_2(k) vs k')
hold on
plot(t,ref21)
plot(t,ref22)

subplot(514)

plot(t,x3_opti_casadi)
ylabel('x_3(k) vs k')
hold on
plot(t,ref31)
plot(t,ref32)

subplot(515)

plot(t,x4_opti_casadi)
ylabel('x_4(k) vs k')
hold on
plot(t,ref41)
plot(t,ref42)

% Ramp-rate plots

figure

subplot(411)

plot(t(2:end),diff1,'r')
ylabel('|x_1(k+1)-x_1(k)| vs k')

subplot(412)

plot(t(2:end),diff2,'g')

```

```

ylabel(' |x_2(k+1)-x_2(k)| vs k ')

subplot(413)

plot(t(2:end), diff3, 'r—')
ylabel(' |x_3(k+1)-x_3(k)| vs k ')

subplot(414)

plot(t(2:end), diff4, 'b')
ylabel(' |x_4(k+1)-x_4(k)| vs k ')

% plot(x1_opti_casadi, c1, 'r--',
%       x2_opti_casadi, c2, 'cO-',
%       x3_opti_casadi, c3, 'b-',
%       x4_opti_casadi, c4, 'k--');

% legend('Gen 1', 'Gen 2', 'Gen 3', 'Gen 4');

```

A.2 Appendix Description

The above code represents the complete computational implementation used in this work. It directly implements the mathematical formulation presented in the previous sections and provides the numerical results used for the analysis.

The program first reads the demand profile from the `demand.xlsx` file and defines the planning horizon. The initial operating points of the four generators are then specified. The CasADi `Opti` framework is subsequently used to define the 44 optimization variables corresponding to the four generator setpoints over the 11 time points.

The nonlinear generation-cost functions are then constructed and combined to form the total objective function. The demand-supply equality, generator operating limits, and ramp-rate constraints are added to the optimization problem before the IPOPT solver is executed.

The optimized generator trajectories are extracted after solving the problem. The code then calculates the total generation and the generation changes between consecutive time steps.

The **first figure** generated by the program compares the calculated total generation with the demand and shows the optimized trajectories of the four generators along with their operating limits.

The **second figure** presents the changes in generator outputs between consecutive time steps, which are used to verify compliance with the 160 MW ramp-rate constraint.

Thus, the MATLAB/CasADi implementation serves as the computational basis for obtaining and verifying the optimal four-generator scheduling solution presented in this report.